INN HOTELS GROUP BUSINESS REPORT

Coded Project

By

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1. Context

A significant number of hotel bookings are called off due to cancellations or no-shows. The typical reasons for cancellations include change of plans, scheduling conflicts, etc. This is often made easier by the option to do so free of charge or preferably at a low cost which is beneficial to hotel guests but it is a less desirable and possibly revenue-diminishing factor for hotels to deal with. Such losses are particularly high on last-minute cancellations.

The new technologies involving online booking channels have dramatically changed customers' booking possibilities and behaviour. This adds a further dimension to the challenge of how hotels handle cancellations, which are no longer limited to traditional booking and guest characteristics.

The cancellation of bookings impacts a hotel on various fronts:

1. Loss of resources (revenue) when the hotel cannot resell the room.
2. Additional costs of distribution channels by increasing commissions or paying for publicity to help sell these rooms.
3. Lowering prices last minute, so the hotel can resell a room, resulting in reducing the profit margin.
4. Human resources to make arrangements for the guests.

2. Objective

The increasing number of cancellations calls for a Machine Learning based solution that can help in predicting which booking is likely to be cancelled. INN Hotels Group has a chain of hotels in Portugal, they are facing problems with the high number of booking cancellations and have reached out to your firm for data-driven solutions. You as a data scientist have to analyse the data provided to find which factors have a high influence on booking cancellations, build a predictive model that can predict which booking is going to be cancelled in advance, and help in formulating profitable policies for cancellations and refunds.

3. Data Description

The data contains the different attributes of customers' booking details. The detailed data dictionary is given below.

Data Dictionary:

- Booking_ID: the unique identifier of each booking
- no_of_adults: Number of adults

- no_of_children: Number of Children
- no_of_weekend_nights: Number of weekend nights (Saturday or Sunday) the guest stayed or booked to stay at the hotel
- no_of_week_nights: Number of weeknights (Monday to Friday) the guest stayed or booked to stay at the hotel
- type_of_meal_plan: Type of meal plan booked by the customer:
    - Not Selected – No meal plan selected
    - Meal Plan 1 – Breakfast
    - Meal Plan 2 – Half board (breakfast and one other meal)
    - Meal Plan 3 – Full board (breakfast, lunch, and dinner)
- required_car_parking_space: Does the customer require a car parking space? (0 - No, 1- Yes)
- room_type_reserved: Type of room reserved by the customer. The values are ciphered (encoded) by INN Hotels Group
- lead_time: Number of days between the date of booking and the arrival date
- arrival_year: Year of arrival date
- arrival_month: Month of arrival date
- arrival_date: Date of the month
- market_segment_type: Market segment designation.
- repeated_guest: Is the customer a repeated guest? (0 - No, 1- Yes)
- no_of_previous_cancellations: Number of previous bookings that were cancelled by the customer prior to the current booking
- no_of_previous_bookings_not_canceled: Number of previous bookings not canceler by the customer prior to the current booking
- avg_price_per_room: Average price per day of the reservation; prices of the rooms are dynamic. (in euros)
- no_of_special_requests: Total number of special requests made by the customer (e.g. high floor, view from the room, etc)
- booking_status: Flag indicating if the booking was cancelled or not.

4. Data Overview through Python

Data which is stored in csv file is imported to Jupyter Notebook with the help of pandas, NumPy libraries and used matplotlib and seaborn libraries to help with data visualization. Dataset is stored in python variable named data.

Understanding the shape of the dataset.

```
(36275, 19)
```

The dataset has 36275 rows and 19 columns.

	Booking_ID	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	type_of_meal_plan	required_car_parking_space	room_type_reserved	lead_time
0	INN00001	2	0	1	2	Meal Plan 1	0	Room_Type 1	224
1	INN00002	2	0	2	3	Not Selected	0	Room_Type 1	5
2	INN00003	1	0	2	1	Meal Plan 1	0	Room_Type 1	1
3	INN00004	2	0	0	2	Meal Plan 1	0	Room_Type 1	211
4	INN00005	2	0	1	1	Not Selected	0	Room_Type 1	48

Figure 1: Top 5 Rows of Dataset

Checking the data types of data.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 36275 entries, 0 to 36274
Data columns (total 19 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Booking_ID                            36275 non-null  object
 1   no_of_adults                          36275 non-null  int64
 2   no_of_children                        36275 non-null  int64
 3   no_of_weekend_nights                  36275 non-null  int64
 4   no_of_week_nights                     36275 non-null  int64
 5   type_of_meal_plan                     36275 non-null  object
 6   required_car_parking_space            36275 non-null  int64
 7   room_type_reserved                    36275 non-null  object
 8   lead_time                             36275 non-null  int64
 9   arrival_year                          36275 non-null  int64
 10  arrival_month                         36275 non-null  int64
 11  arrival_date                          36275 non-null  int64
 12  market_segment_type                   36275 non-null  object
 13  repeated_guest                        36275 non-null  int64
 14  no_of_previous_cancellations          36275 non-null  int64
 15  no_of_previous_bookings_not_canceled  36275 non-null  int64
 16  avg_price_per_room                    36275 non-null  float64
 17  no_of_special_requests                36275 non-null  int64
 18  booking_status                        36275 non-null  object
dtypes: float64(1), int64(13), object(5)
memory usage: 5.3+ MB
```

- The `Booking ID`, `type_of_meal_plan`, `room_type_reserved`, `market_segment_type`, `booking_status` columns are of *object* type while the rest columns are numeric in nature

Figure 2: DataTypes in the dataset

4.1 Checking for missing values in the data.

```
Booking_ID                            0
no_of_adults                          0
no_of_children                        0
no_of_weekend_nights                  0
no_of_week_nights                     0
type_of_meal_plan                     0
required_car_parking_space            0
room_type_reserved                    0
lead_time                             0
arrival_year                          0
arrival_month                         0
arrival_date                          0
market_segment_type                   0
repeated_guest                        0
no_of_previous_cancellations          0
no_of_previous_bookings_not_canceled  0
avg_price_per_room                    0
no_of_special_requests                0
booking_status                        0
dtype: int64
```

- There are no null values in the dataset

Figure 3: Missing values in Attributes

4.2 Checking for duplicate values in the data.

```
0
```

- There are no duplicate values in the data.

Figure 4: Duplicate values check

4.3 Checking the statistical summary of the numerical data.

	count	mean	std	min	25%	50%	75%	max
no_of_adults	36275.00000	1.84496	0.51871	0.00000	2.00000	2.00000	2.00000	4.00000
no_of_children	36275.00000	0.10528	0.40265	0.00000	0.00000	0.00000	0.00000	10.00000
no_of_weekend_nights	36275.00000	0.81072	0.87064	0.00000	0.00000	1.00000	2.00000	7.00000
no_of_week_nights	36275.00000	2.20430	1.41090	0.00000	1.00000	2.00000	3.00000	17.00000
required_car_parking_space	36275.00000	0.03099	0.17328	0.00000	0.00000	0.00000	0.00000	1.00000
lead_time	36275.00000	85.23256	85.93082	0.00000	17.00000	57.00000	126.00000	443.00000
arrival_year	36275.00000	2017.82043	0.38384	2017.00000	2018.00000	2018.00000	2018.00000	2018.00000
arrival_month	36275.00000	7.42365	3.06989	1.00000	5.00000	8.00000	10.00000	12.00000
arrival_date	36275.00000	15.59700	8.74045	1.00000	8.00000	16.00000	23.00000	31.00000
repeated_guest	36275.00000	0.02564	0.15805	0.00000	0.00000	0.00000	0.00000	1.00000
no_of_previous_cancellations	36275.00000	0.02335	0.36833	0.00000	0.00000	0.00000	0.00000	13.00000
no_of_previous_bookings_not_canceled	36275.00000	0.15341	1.75417	0.00000	0.00000	0.00000	0.00000	58.00000
avg_price_per_room	36275.00000	103.42354	35.08942	0.00000	80.30000	99.45000	120.00000	540.00000
no_of_special_requests	36275.00000	0.61966	0.78624	0.00000	0.00000	0.00000	1.00000	5.00000

Figure 5: Statistical Summary of the numerical data

1. Number of Adults

The range varies from a minimum of 0 adults to a maximum of 4 adults per booking. Most bookings (25%, 50%, and 75%) typically involve 2 adults.

2. Number of Children

The number of children ranges from 0 to 10. A significant majority of bookings (25%, 50%, and 75%) do not include any children.

3. Number of Weekend Nights

The number of weekend nights ranges from 0 to 7. Many bookings (25%) have no weekend nights, while 50% and 75% of bookings have 1 and 2 weekend nights, respectively.

4. Number of Week Nights

The number of week nights ranges from 0 to 17. While 25% of bookings include 1 week night, 50% have 2-week nights, and 75% have 3-week nights.

5. Required Car Parking Space

The number of required parking spaces ranges from 0 to 1, with the majority of bookings (25%, 50%, and 75%) not requiring any car parking space.

6. Lead Time

The lead time ranges from 0 to 443 days. While 25% of bookings have a lead time of 17 days, 50% have a lead time of 57 days, and 75% have a lead time of 126 days.

7. Arrival Year

The arrival years range from 2017 to 2018, with 25% and 75% of bookings recorded in 2017 and 50% of bookings in 2018.

8. Arrival Month

The arrival months range from January (1) to December (12). The months of May, August, and October represent the 25th, 50th, and 75th percentiles, respectively.

9. Arrival Date

The arrival dates range from the 1st to the 31st of the month. The 8th, 16th, and 23rd days of the month represent the 25th, 50th, and 75th percentiles, respectively.

10. Repeated Guest

The number of repeated guests ranges from 0 to 1, with the majority of bookings (25%, 50%, and 75%) not involving repeated guests.

11. Number of Previous Cancellations

The number of previous cancellations ranges from 0 to 13. Most bookings (25%, 50%, and 75%) have no previous cancellations.

12. Number of Previous Bookings Not Cancelled

The number of previous bookings not cancelled ranges from 0 to 72. A significant majority of bookings (25%, 50%, and 75%) have no previous bookings not cancelled.

13. Average Price per Room

The price per room ranges from 0 to 540 units. The prices at the 25th, 50th, and 75th percentiles are 69.29, 94.50, and 126.00 units, respectively.

14. Number of Special Requests

The number of special requests ranges from 0 to 5. Many bookings (25%) have no special requests, while 50% and 75% of bookings have 0 and 1 special request, respectively.

5.0 Univariate Analysis

5.1 Observations on no_of_adults

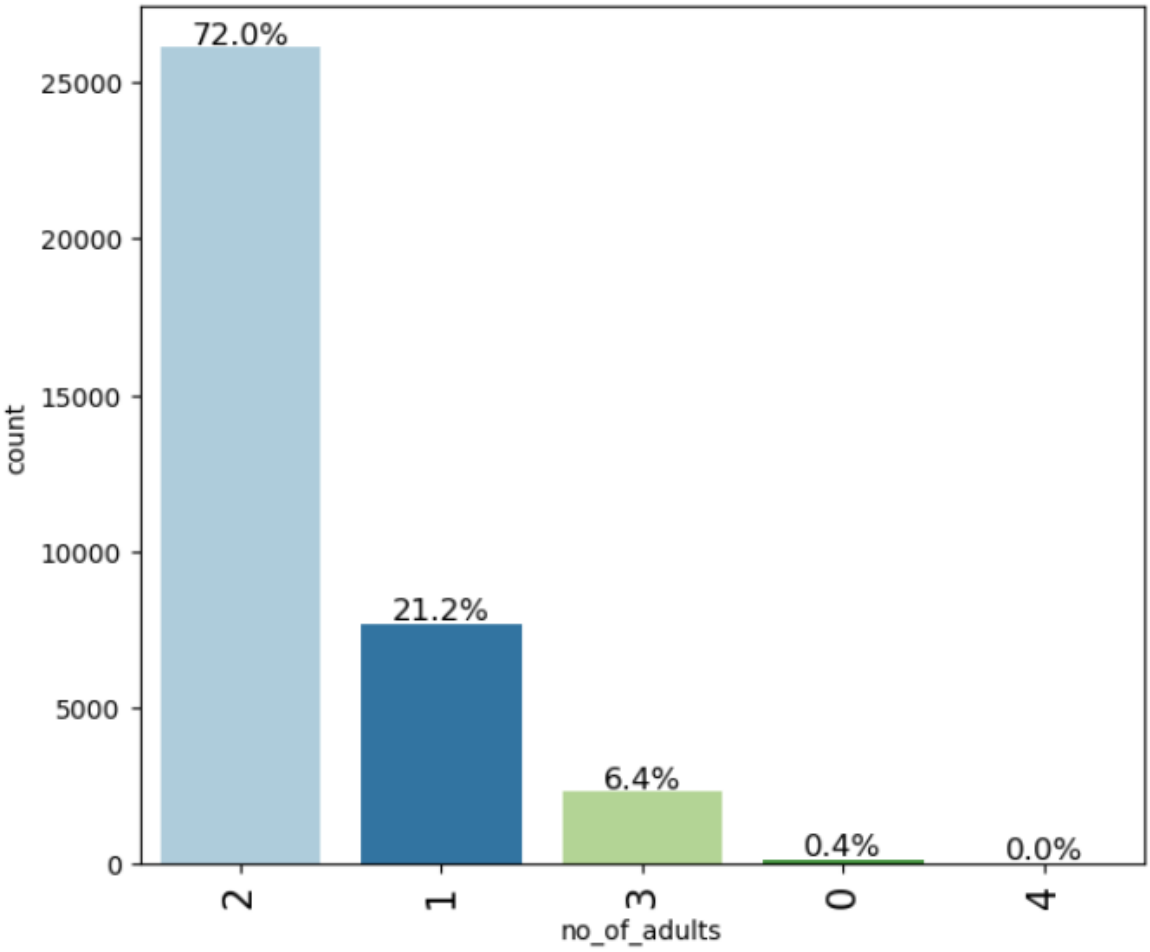


Figure 6: Observations on No.of adults

Around 72% of are of 2 adults, 21% are single adults whereas 6.4% are of 3 adults.

5.2 Observations on no_of_children

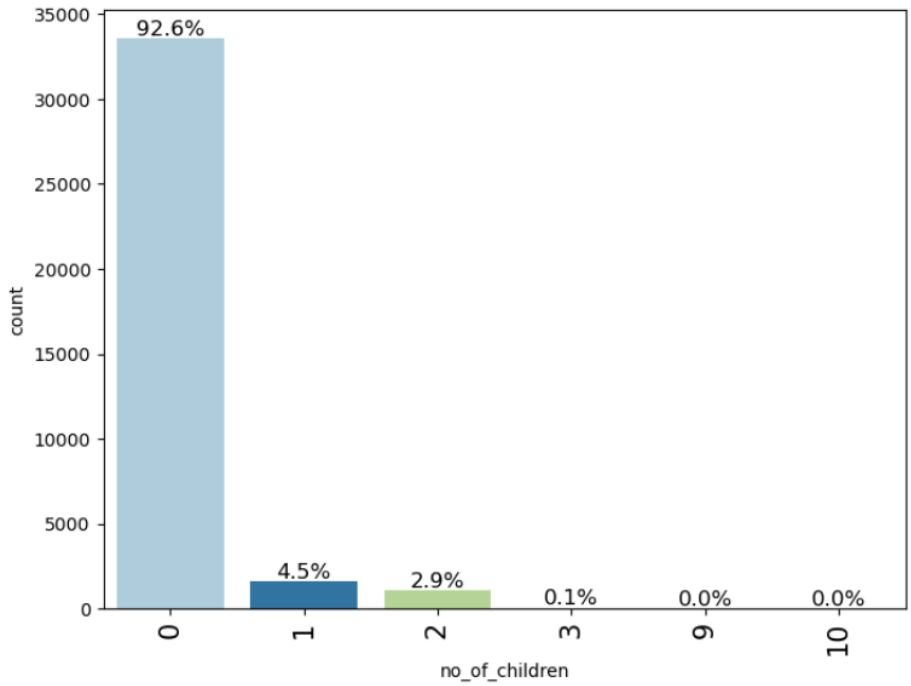


Figure 7: Observations on No of Children

- Around 92.6% of bookings are of zero children and 4.5% are of 1 child and 2.9% are of 2 children

5.3 Observations on no_of_weekend_nights

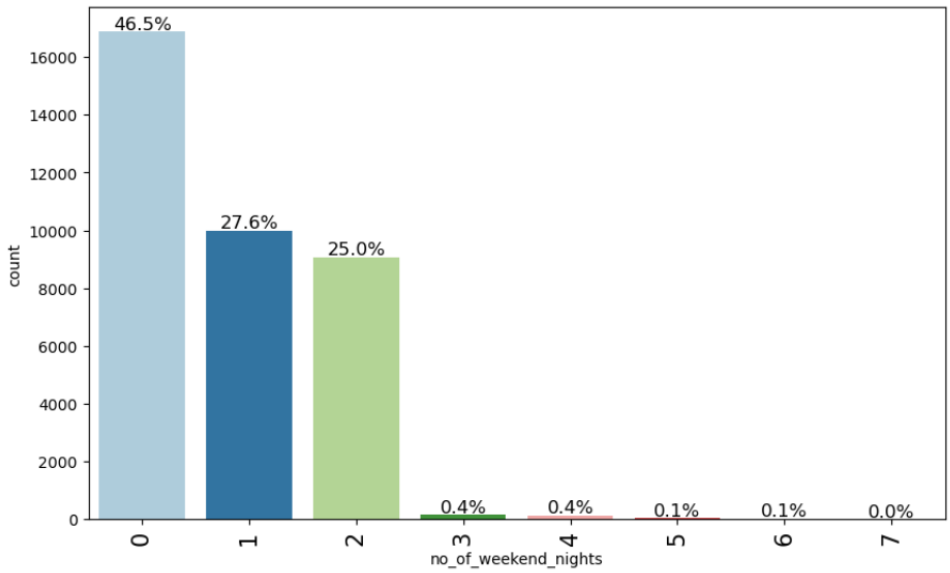


Figure 8: Observations on no_of_weekend_nights

- Most of the bookings are of 0 weekend nights and 1 and 2 stands next respectively

5.4 Observations on no_of_week_nights

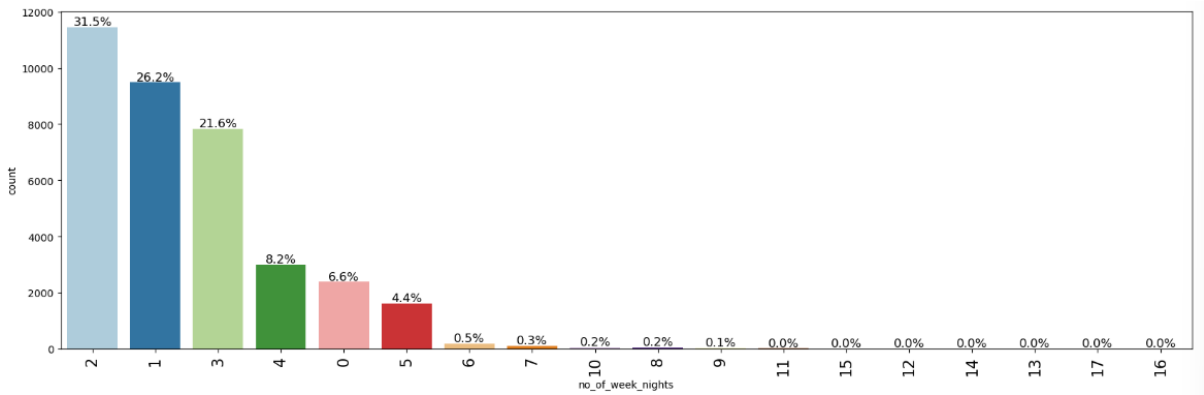


Figure 9: Observations on no_of_week_nights

- Most of the booking contains 2 weeknights at least. next is 1 and 3 and 4

5.5 Observations on required_car_parking_space

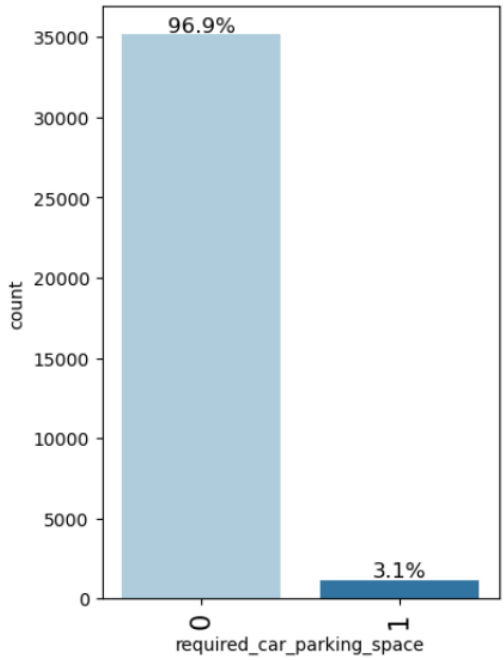


Figure 10: Observations on required_car_parking_space

- Most of the bookings 96.9% does not required car parking space.

5.6 Observations on lead_time

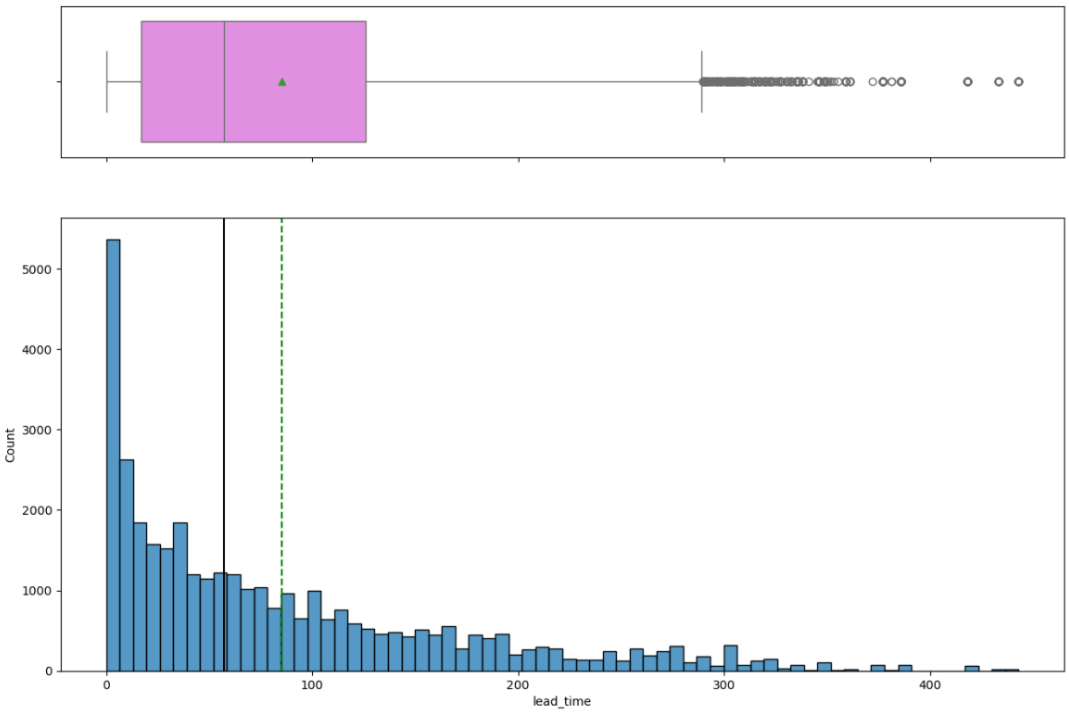


Figure 11: Observations on lead_time

- The distribution of lead_time is uniformly distributed with some higher values being less frequent.
- Outliers are present in this attribute.

5.7 Observations on arrival_year

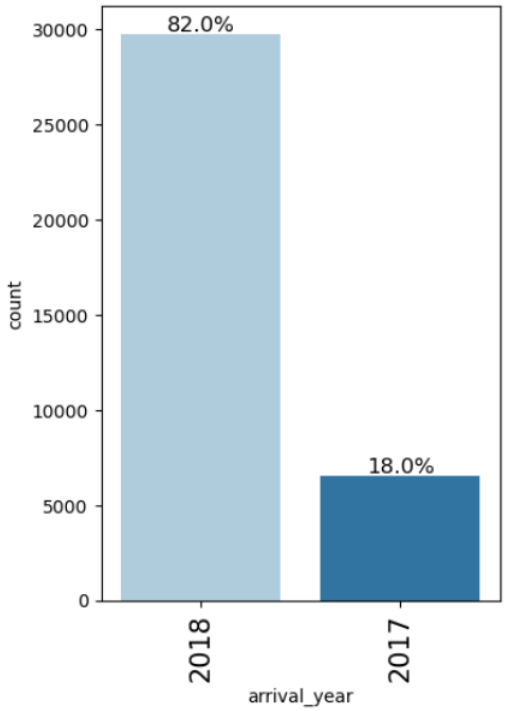


Figure 12: Observations on arrival_year

- Most of the bookings 82.0% are of year 2018

5.8 Observations on arrival_month

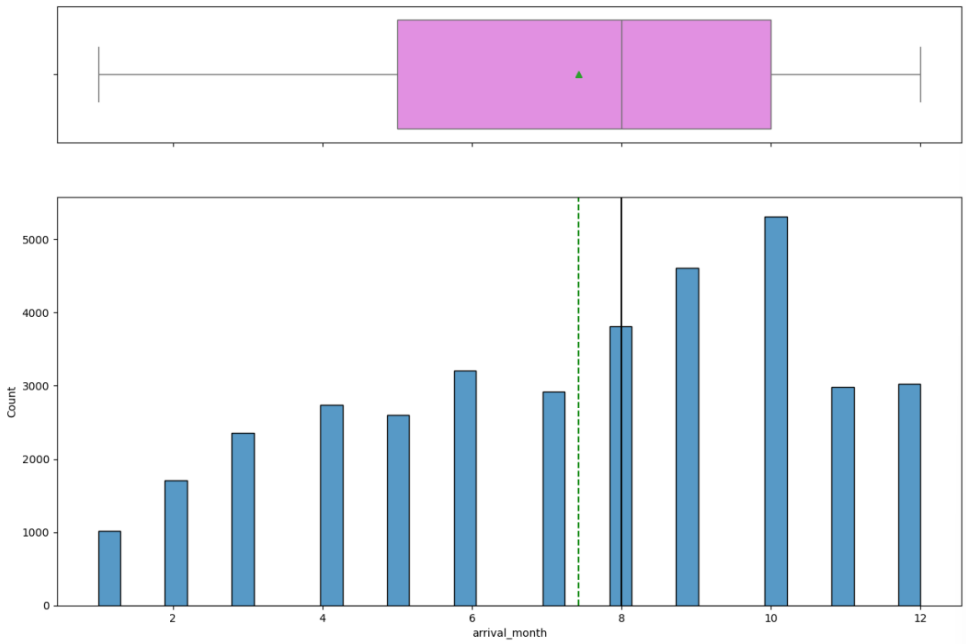


Figure 13: Observations on arrival_month

- Most of the bookings are in Month of October, September, and August.

5.9 Observations on arrival_date

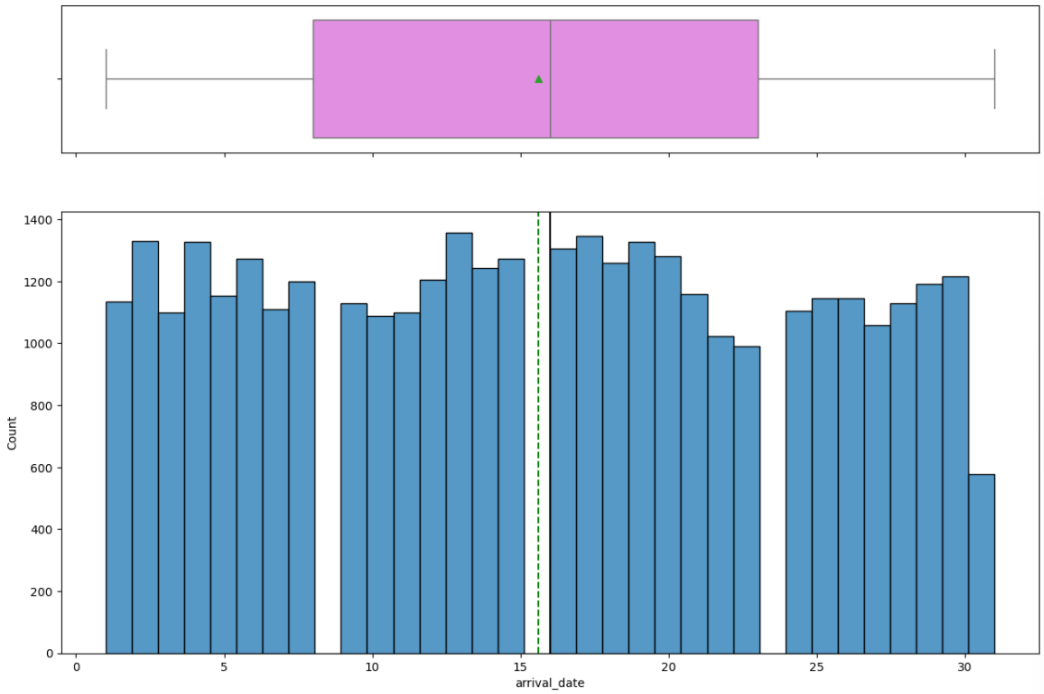


Figure 14: Observations on arrival_date

- The distribution of arrival_date is uniformly distributed
- No Outliers are present.

5.10 Observations on repeated_guest

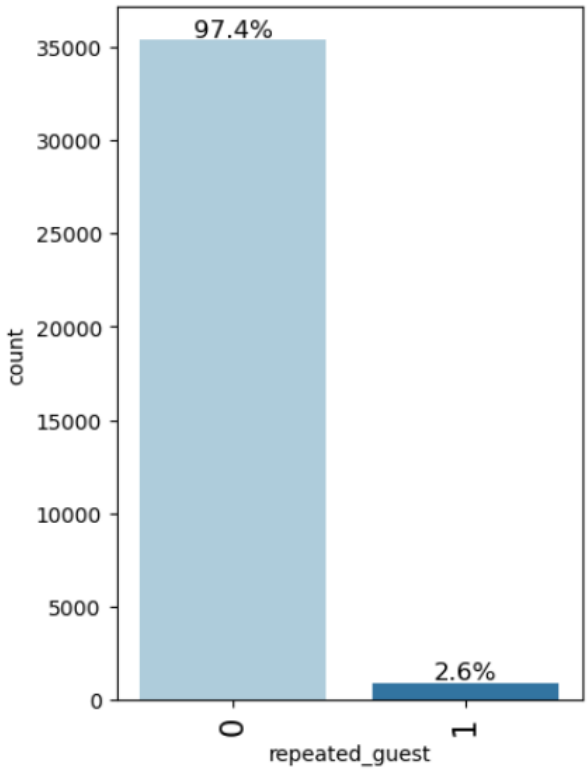


Figure 15: Observations on repeated_guest

- Most of the Bookings are First timers and and repeating guest are < 3%

5.11 Observations on no_of_previous_cancellations

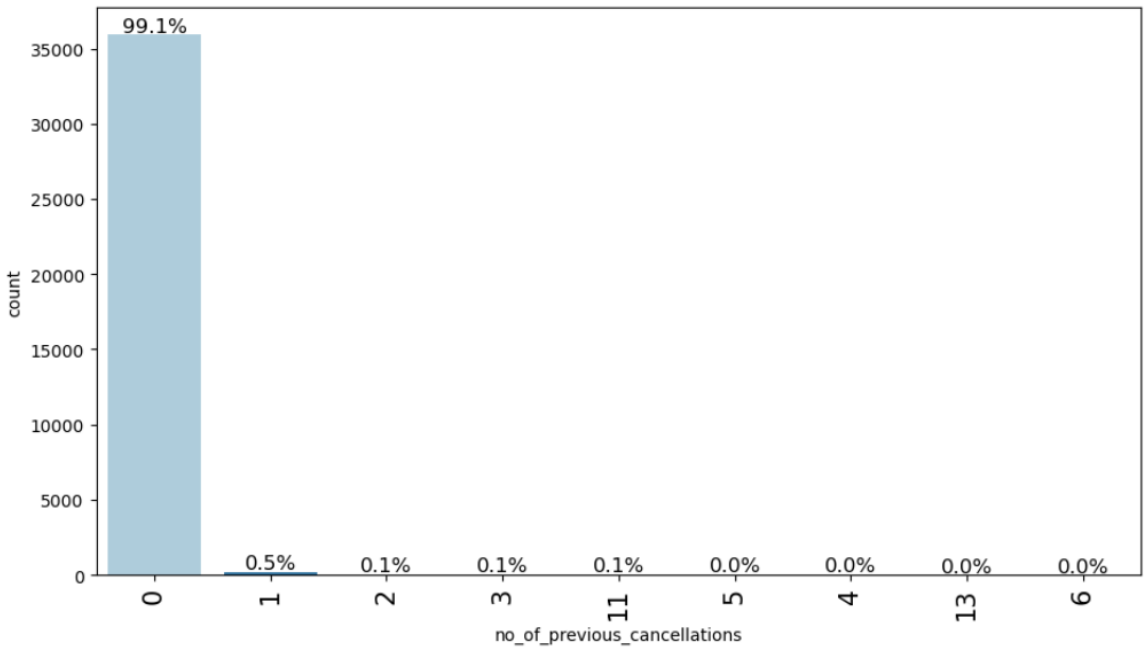


Figure 16: Observations on no_of_previous_cancellations

- Most of the Bookings 99.1% have not cancelled bookings previously.

5.12 Observations on no_of_previous_bookings_not_canceled

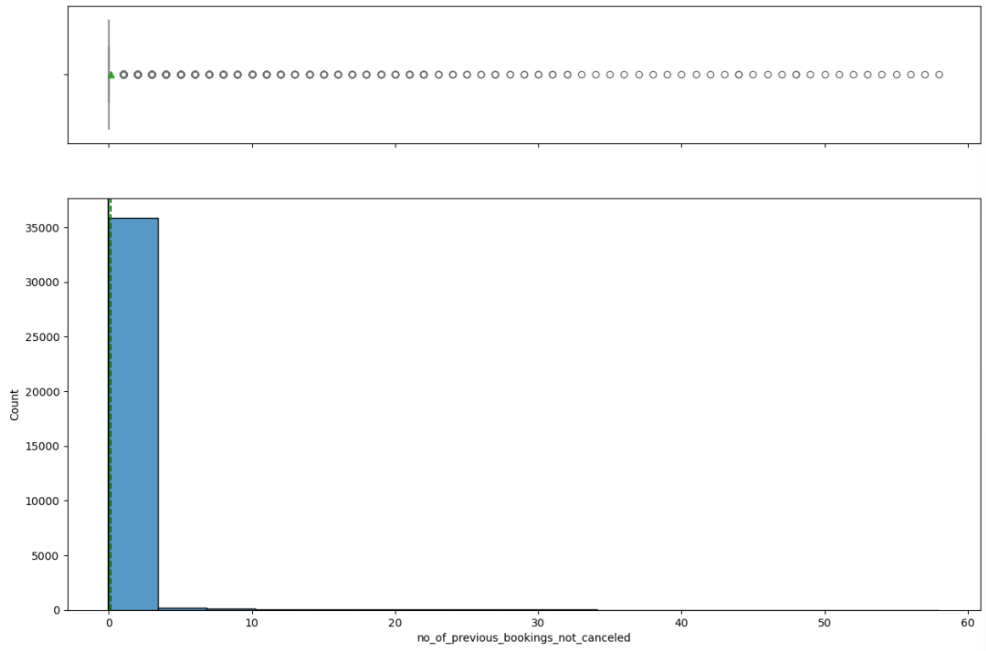


Figure 17: Observations on no_of_previous_bookings_not_canceled

- Most of the Bookings are not cancelled bookings previously.

5.13 Observations on avg_price_per_room

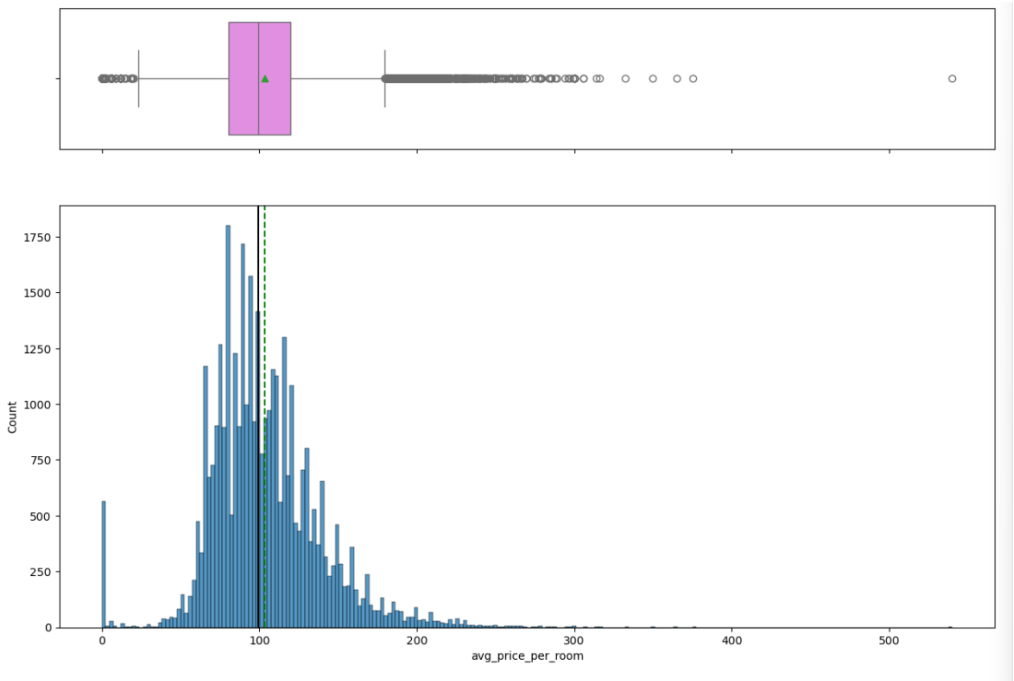


Figure 18: Observations on avg_price_per_room

- avg_price_per_room attribute data is normally distributed

- There are outliers in this attribute.

5.14 Observations on market_segment_type

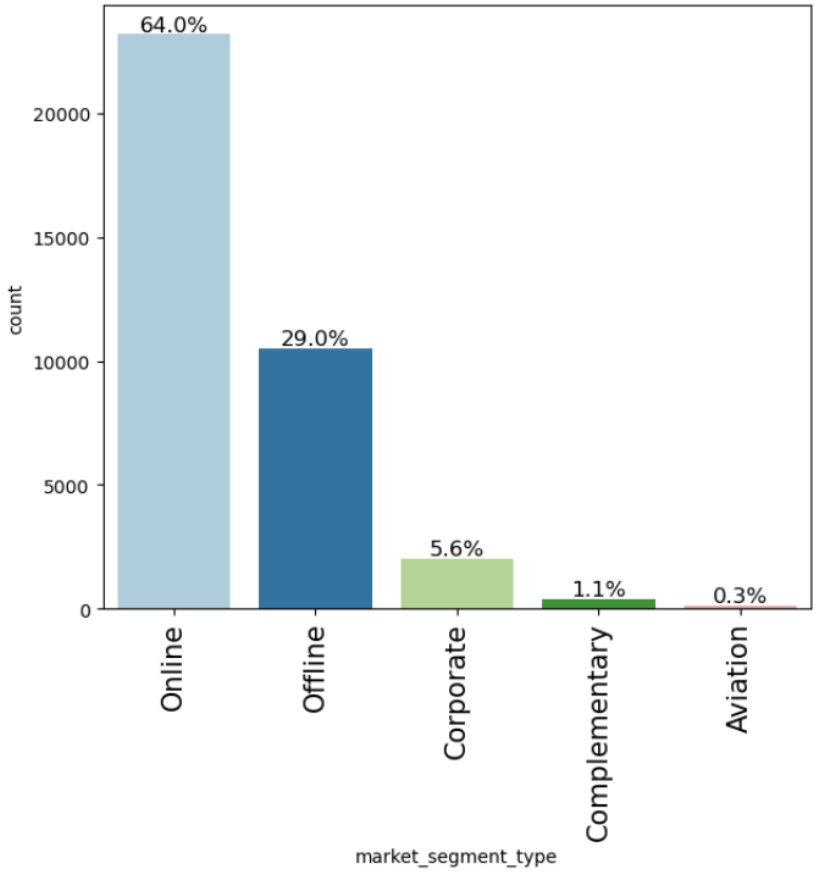


Figure 19: Observations on market_segment_type

Most of the bookings are of Online, then Offline.

5.15 Observations on no_of_special_requests

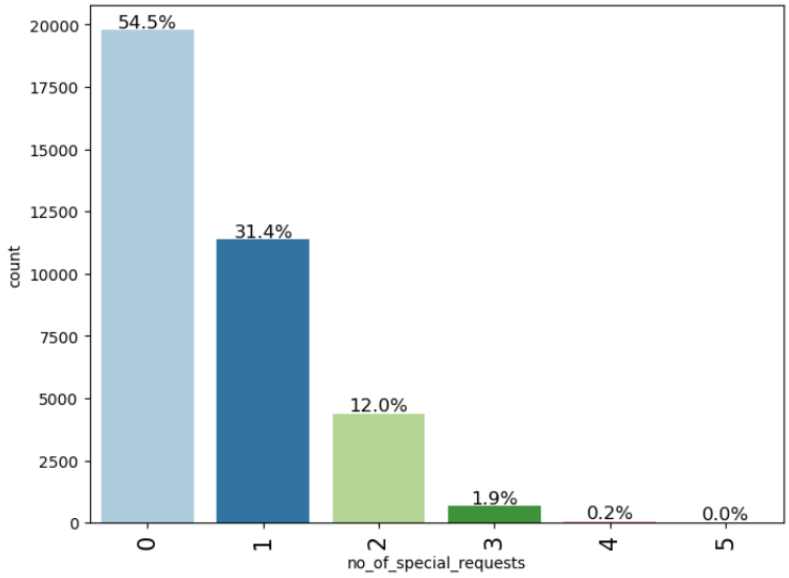


Figure 20: Observations on no_of_special_requests

- Most of the bookings are not having special requests

5.16 Observations on booking_status

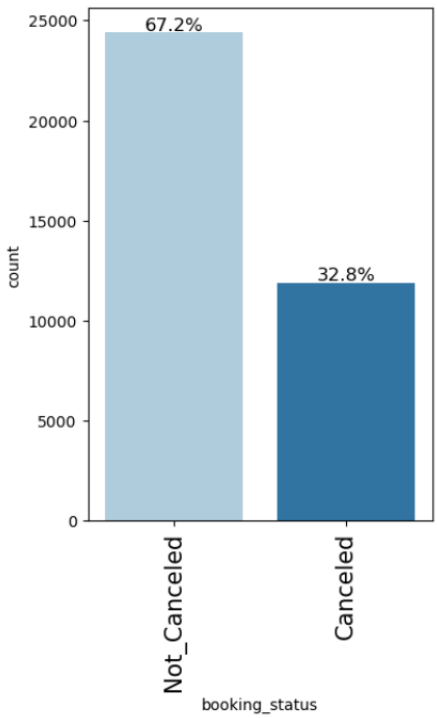


Figure 21: Observations on booking_status

- 32.8% of bookings are cancelled.

6.0 Bivariate Analysis

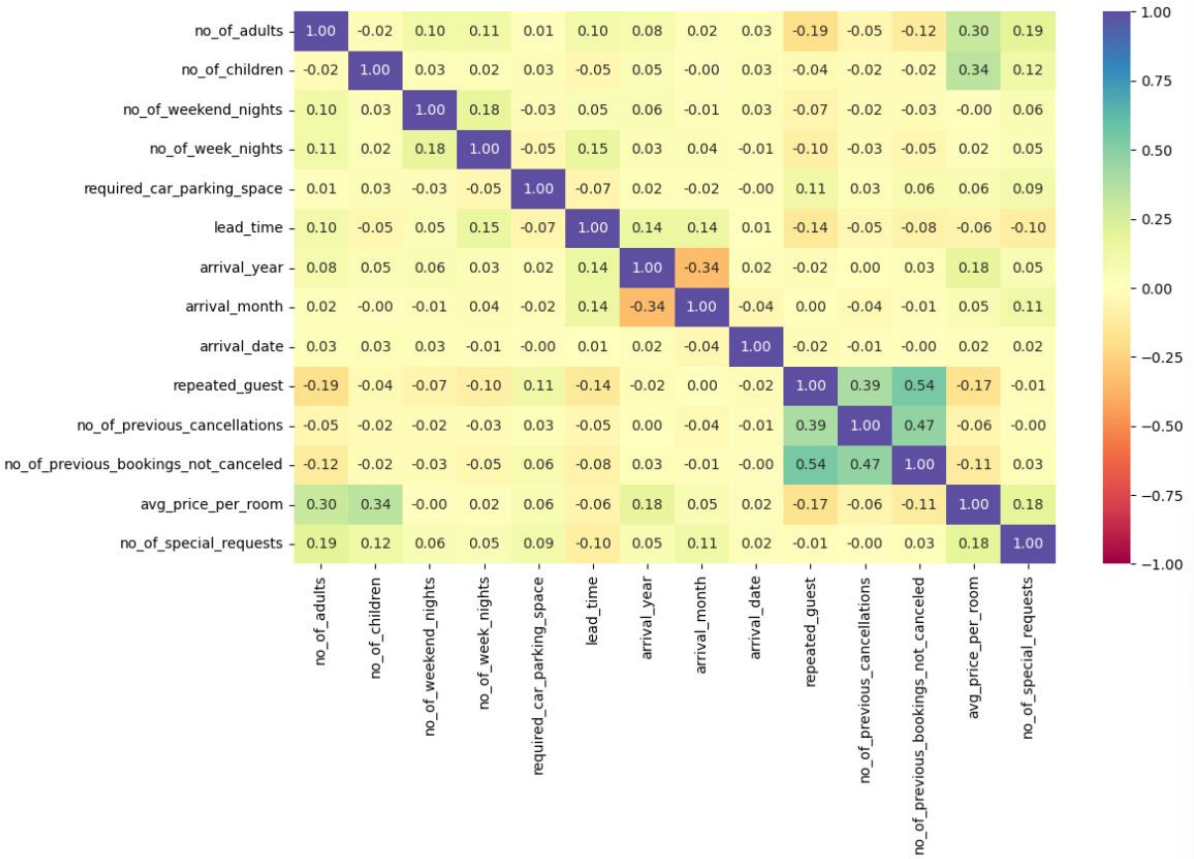


Figure 22: Correlation heat map b/w variables

- Lead Time vs. Average Price per Room: There is a moderate positive correlation between lead time and the average price per room. This suggests that bookings made further in advance tend to have higher average room prices.
- Number of Previous Cancellations vs. Lead Time: The number of previous cancellations shows a slightly positive correlation with lead time. This could indicate that guests who tend to cancel their bookings also book further in advance.
- Number of Special Requests vs. Average Price per Room: There is a slight positive correlation between the number of special requests and the average price per room. Higher spending guests might demand more special requests.
- Weekend Nights vs. Week Nights: There's a moderate positive correlation between the number of weekend nights and week nights, which could imply that guests who stay over the weekend are more likely to extend their stay into the weekdays as well.
- Number of Children vs. Number of Adults: The number of children shows a moderate positive correlation with the number of adults, which is expected as larger groups or families tend to book together.

- Required Car Parking Space vs. Number of Special Requests: There is a slight positive correlation between required car parking space and the number of special requests, indicating that guests who need parking may also have more specific needs.
- Repeated Guest: Repeated guest status shows weak correlations with most variables, suggesting that repeated bookings do not heavily influence most of the other variables like special requests or average price per room.
- Lead Time vs. Number of Special Requests: There is a slight positive correlation between lead time and the number of special requests, indicating that guests who plan further ahead are likely to have more specific needs or requests.

6.1 Observations on booking_status vs no_of_special_requests

booking_status	Canceled	Not_Canceled	All
no_of_special_requests			
All	11885	24390	36275
0	8545	11232	19777
1	2703	8670	11373
2	637	3727	4364
3	0	675	675
4	0	78	78
5	0	8	8

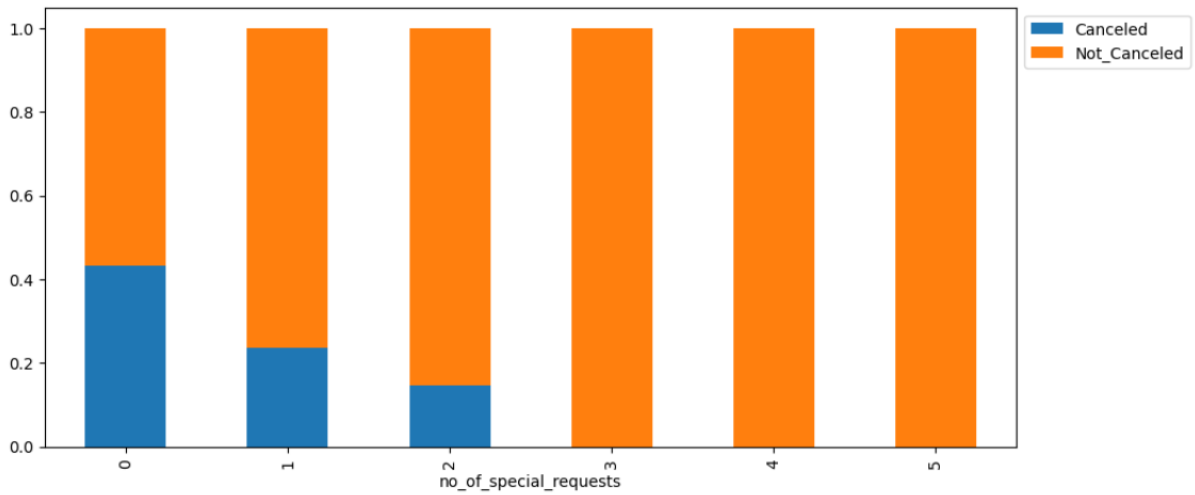


Figure 23: Observations on booking_status vs no_of_special_requests

Most of the bookings are cancelled from customers who provided 0 requests and 1 and 2 requests

6.2 Observations on booking_status vs no_of_previous_cancellations

booking_status	Canceled	Not_Canceled	All
no_of_previous_cancellations			
All	11885	24390	36275
0	11869	24068	35937
1	11	187	198
13	4	0	4
3	1	42	43
2	0	46	46
4	0	10	10
5	0	11	11
6	0	1	1
11	0	25	25

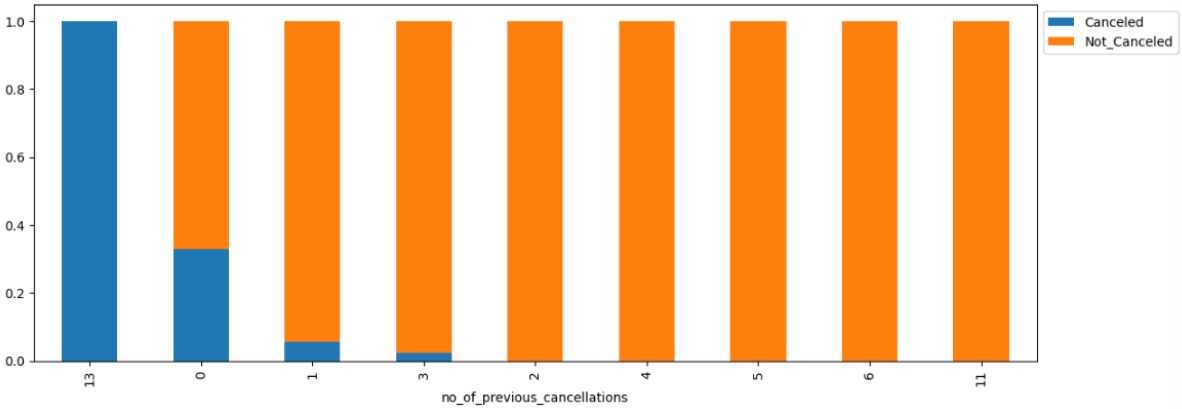


Figure 24: Observations on booking_status vs no_of_previous_cancellations

Most of the cancellations are from customers who were first time customers.

6.3 Observations on booking_status vs avg_price_per_room

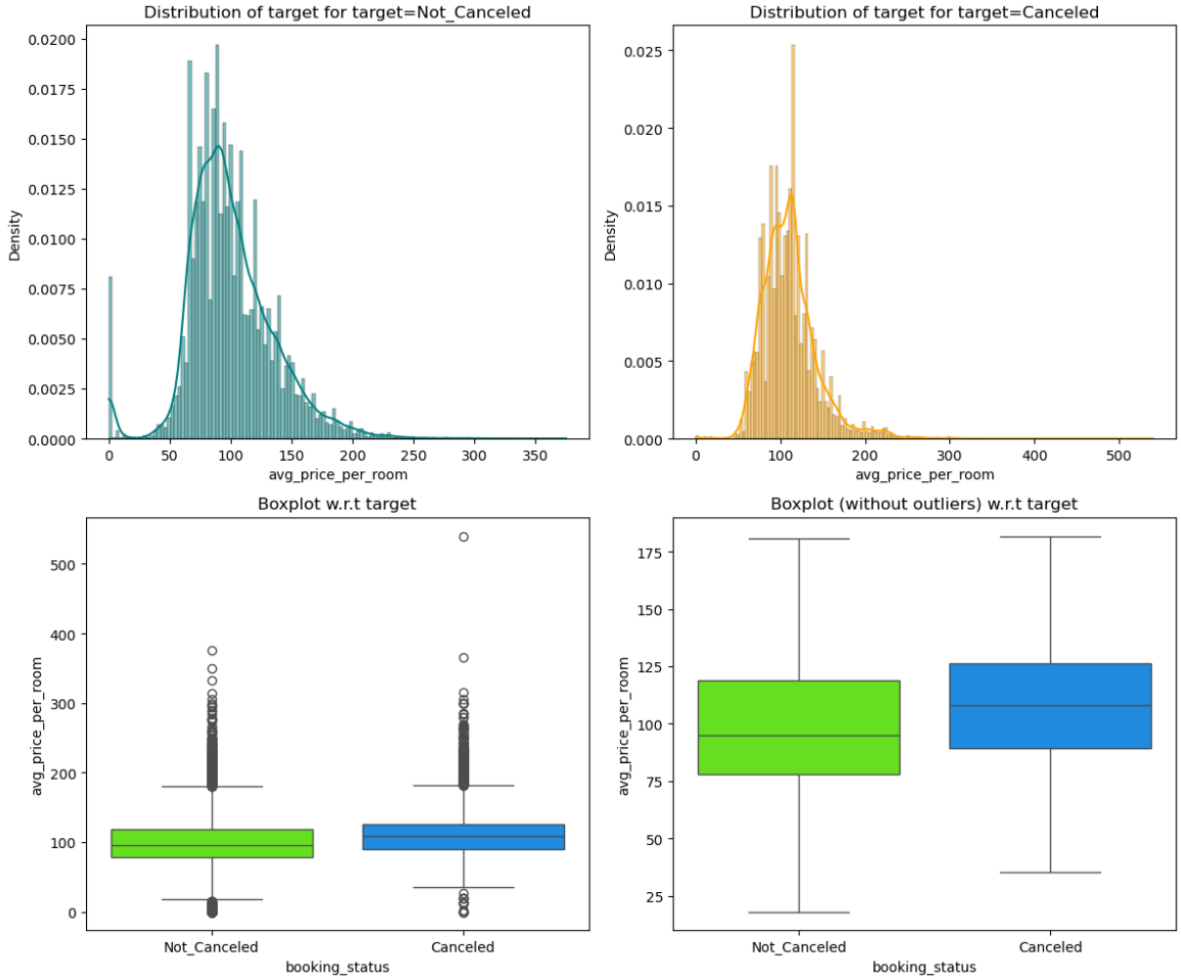


Figure 25: Observations on booking_status vs avg_price_per_room

- Bookings with an average room price between 50 and 100 are less likely to be cancelled.
- There is a broader range of prices for cancelled bookings, with a notable density between 60 and 80.
- The median average price per room is higher for not-cancelled bookings compared to cancelled ones.

6.4 Observations on booking_status vs market_segment_type

booking_status	Canceled	Not_Canceled	All
market_segment_type			
All	11885	24390	36275
Online	8475	14739	23214
Offline	3153	7375	10528
Corporate	220	1797	2017
Aviation	37	88	125
Complementary	0	391	391

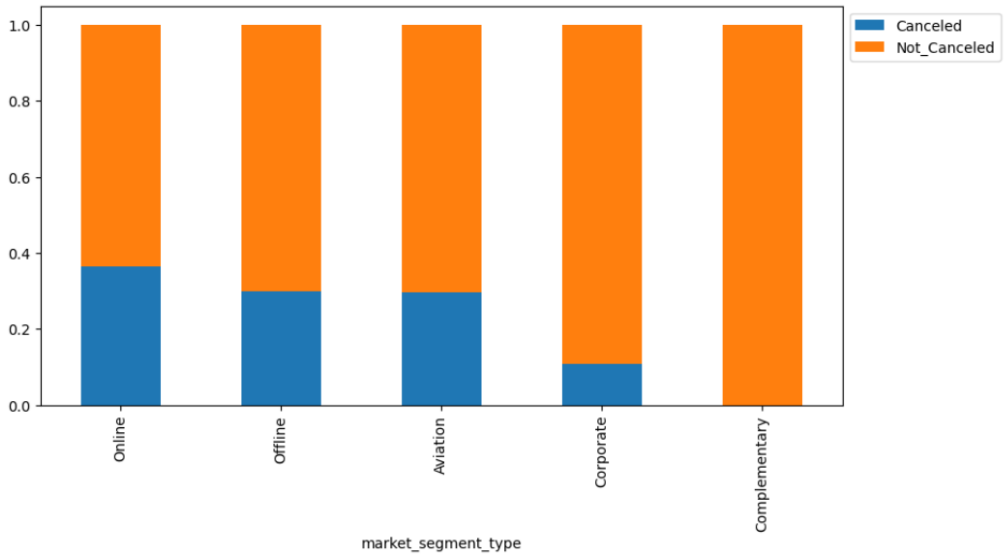


Figure 26: Observations on booking_status vs market_segment_type

- The Corporate and Offline segments have the highest proportions of not canceled bookings, indicating that these segments are more stable in terms of booking retention.
- The Online segment has the highest number of canceled bookings in absolute terms and also has a relatively high proportion of cancellations.
- The Complementary segment has no cancellations, which might suggest either a small sample size or very specific conditions that ensure bookings are not canceled.

6.5 Observations on booking_status vs lead_time

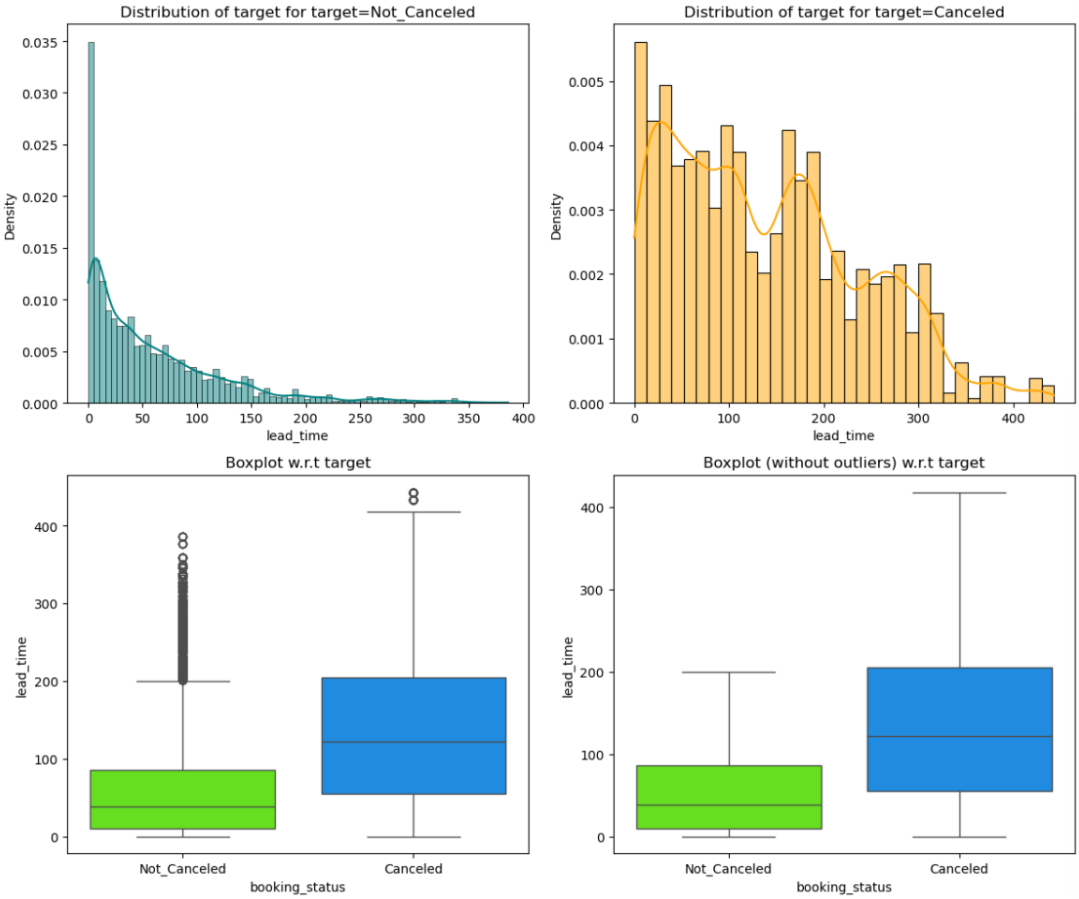


Figure 27: Observations on booking_status vs lead_time

- Shorter lead times are associated with a lower likelihood of cancellation.
- Longer lead times are associated with a higher likelihood of cancellation.

6.6 Observations on booking_status vs room_type_reserved

booking_status	Canceled	Not_Canceled	All
room_type_reserved			
All	11885	24390	36275
Room_Type 1	9072	19058	28130
Room_Type 4	2069	3988	6057
Room_Type 6	406	560	966
Room_Type 2	228	464	692
Room_Type 5	72	193	265
Room_Type 7	36	122	158
Room_Type 3	2	5	7

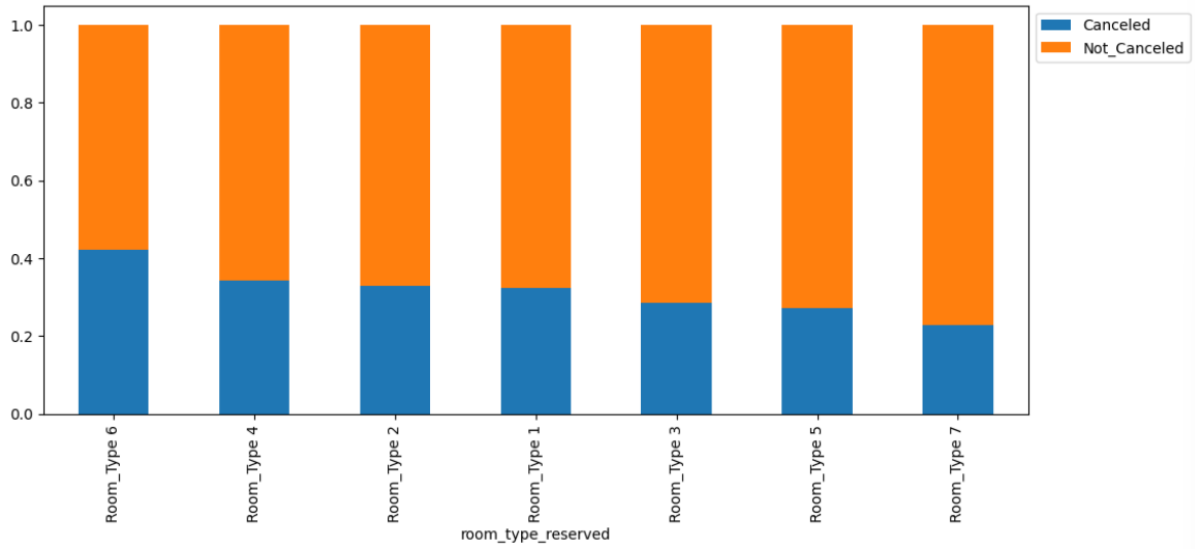


Figure 28: Observations on booking_status vs room_type_reserved

- Room_Type 1 has the highest number of total bookings and the highest number of not canceled bookings, suggesting it is the most popular room type.
- Room_Type 6 has the highest proportion of cancelled bookings, indicating a higher likelihood of cancellations for this room type.
- Room_Type 7 has the lowest proportion of cancelled bookings, suggesting higher customer satisfaction or better retention strategies for this room type.

6.7 Observations on booking_status vs repeated_guest

```
booking_status  Canceled  Not_Canceled    All
repeated_guest
All                11885         24390  36275
0                  11869         23476  35345
1                     16           914    930
```

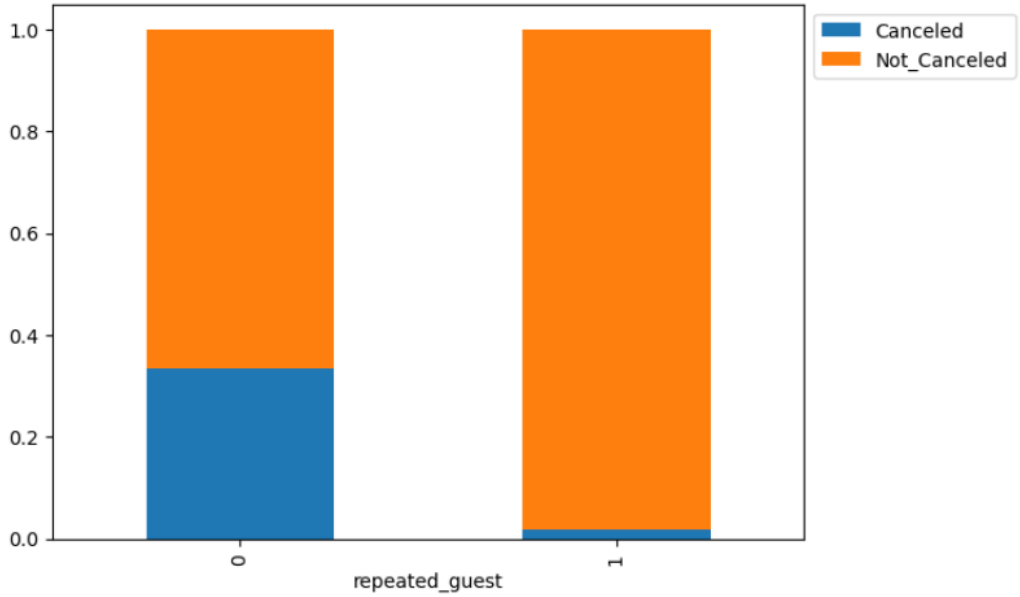


Figure 29: Observations on booking_status vs repeated_guest

- A significantly higher proportion of bookings are not cancelled for repeated guests compared to non-repeated guests.
- The overall cancellation rate is higher among non-repeated guests.

6.8 Observations on booking_status vs required_car_parking_space

```
booking_status              Canceled  Not_Canceled    All
required_car_parking_space
All                            11885         24390  36275
0                              11771         23380  35151
1                                114          1010   1124
```

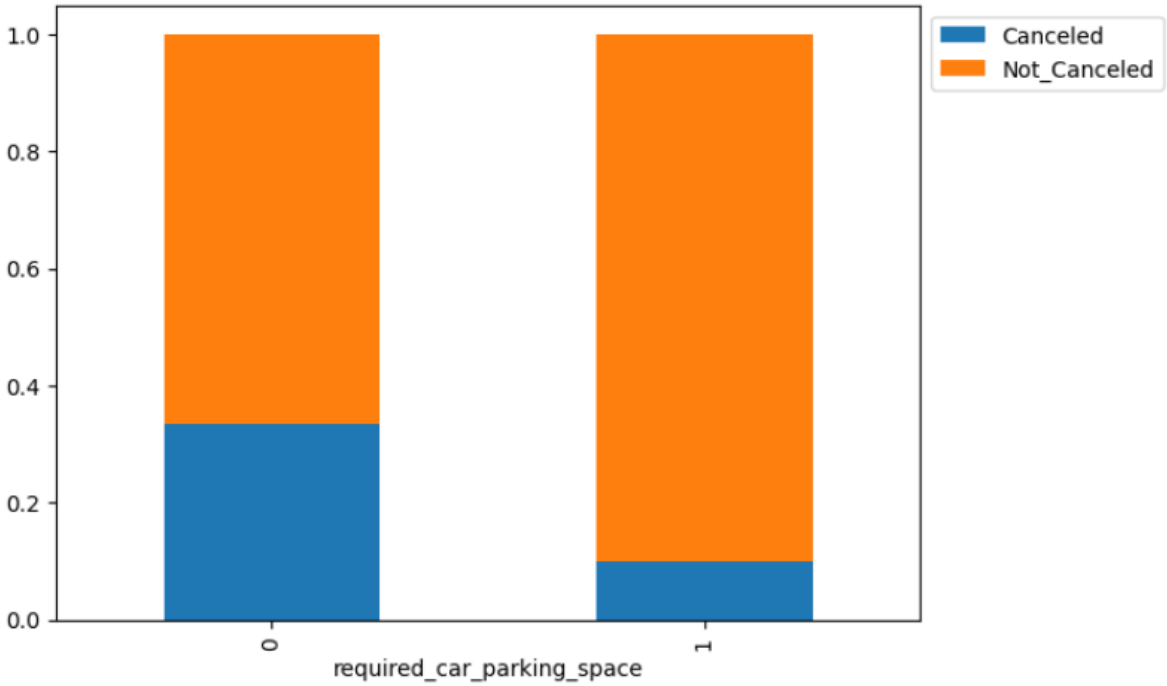


Figure 30: Observations on booking_status vs required_car_parking_space

- Bookings that required car parking space (1) have a significantly lower cancellation rate (approximately 10.1%) compared to those that did not require car parking space (approximately 33.5%).
- This suggests that customers who need car parking space are more committed to their bookings, possibly due to additional travel plans or requirements.

6.9 Observations on booking_status vs type_of_meal_plan

```
booking_status     Canceled  Not_Canceled    All
type_of_meal_plan
All                  11885         24390  36275
Meal Plan 1           8679         19156  27835
Not Selected          1699          3431   5130
Meal Plan 2           1506          1799   3305
Meal Plan 3              1             4      5
```

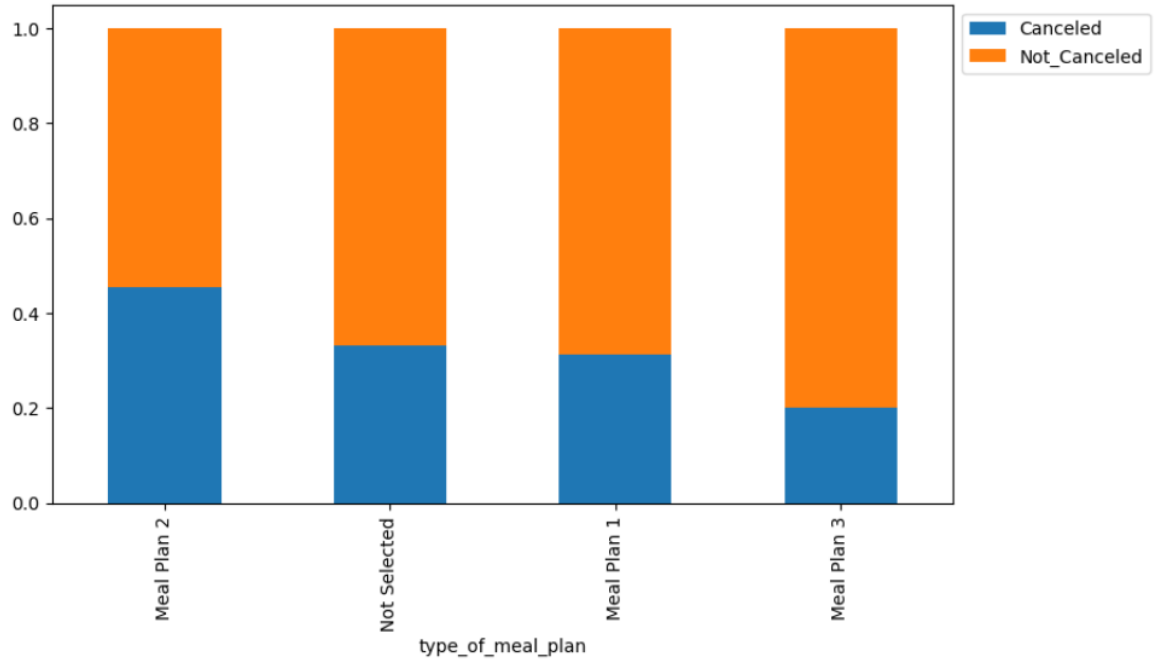


Figure 31: Observations on booking_status vs type_of_meal_plan

- Meal Plan 1 has the highest number of total bookings, both cancelled and not cancelled, suggesting it is the most popular meal plan.
- Meal Plan 3 has the lowest number of total bookings but has a high proportion of not cancelled bookings.
- The highest cancellation rate is seen in Meal Plan 2, with nearly half of the bookings being cancelled.
- The proportion of cancelled bookings is relatively consistent for Meal Plan 1 and bookings with no meal plan selected.

6.10 Observations on avg_price_per_room vs market_segment_type

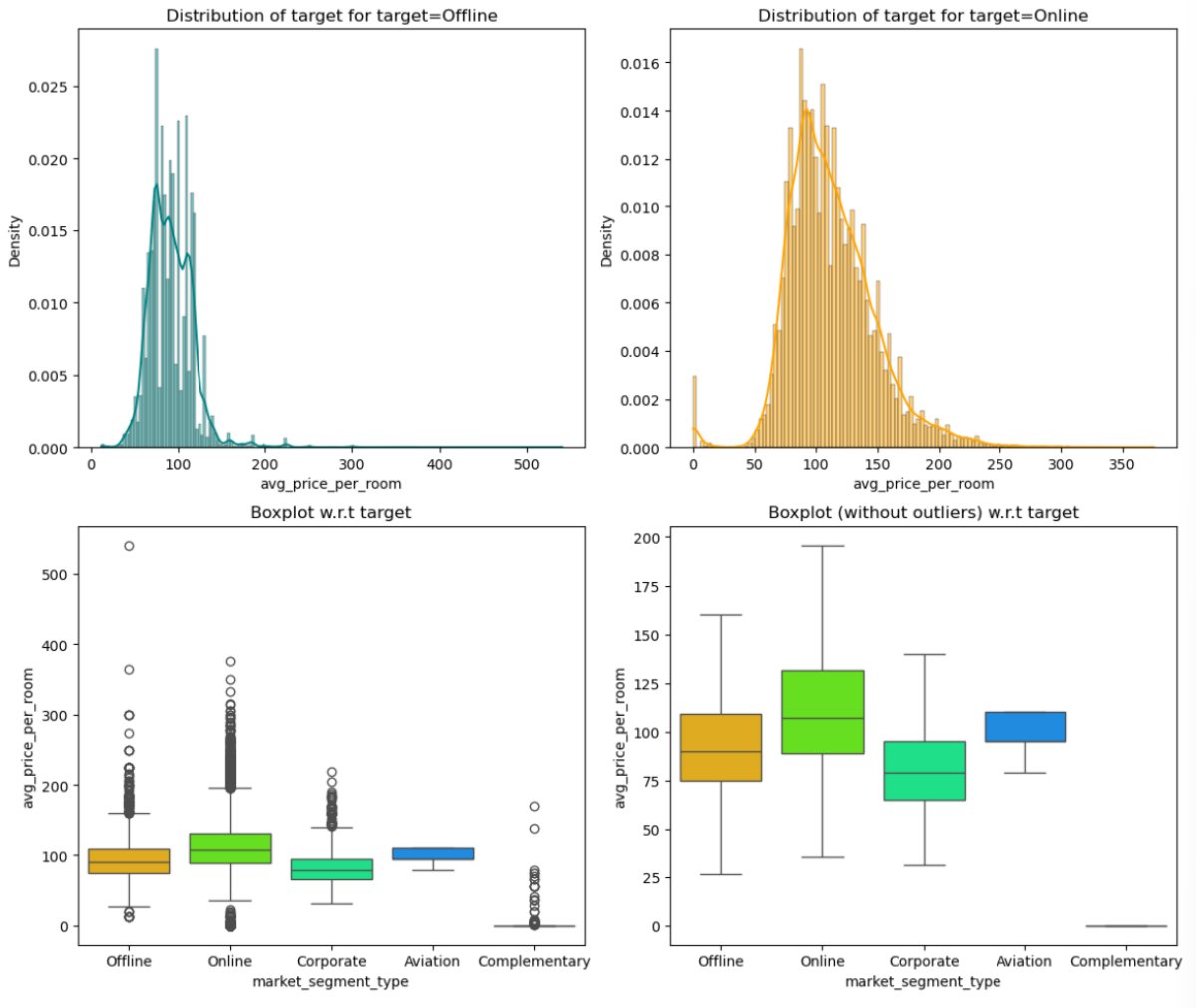


Figure 32: Observations on avg_price_per_room vs market_segment_type

- Offline bookings have a median price of 75, with most prices falling between 50 to 100.
- Online bookings have a slightly higher median price of 80, with most prices between 55 to 105.
- Corporate bookings have the highest median price of 90, indicating that this segment is willing to pay more.
- Aviation and Complementary bookings show different price ranges, with Complementary bookings having the lowest median price of 65.

7.0 Finding Outliers

Finding outliers on numerical columns.

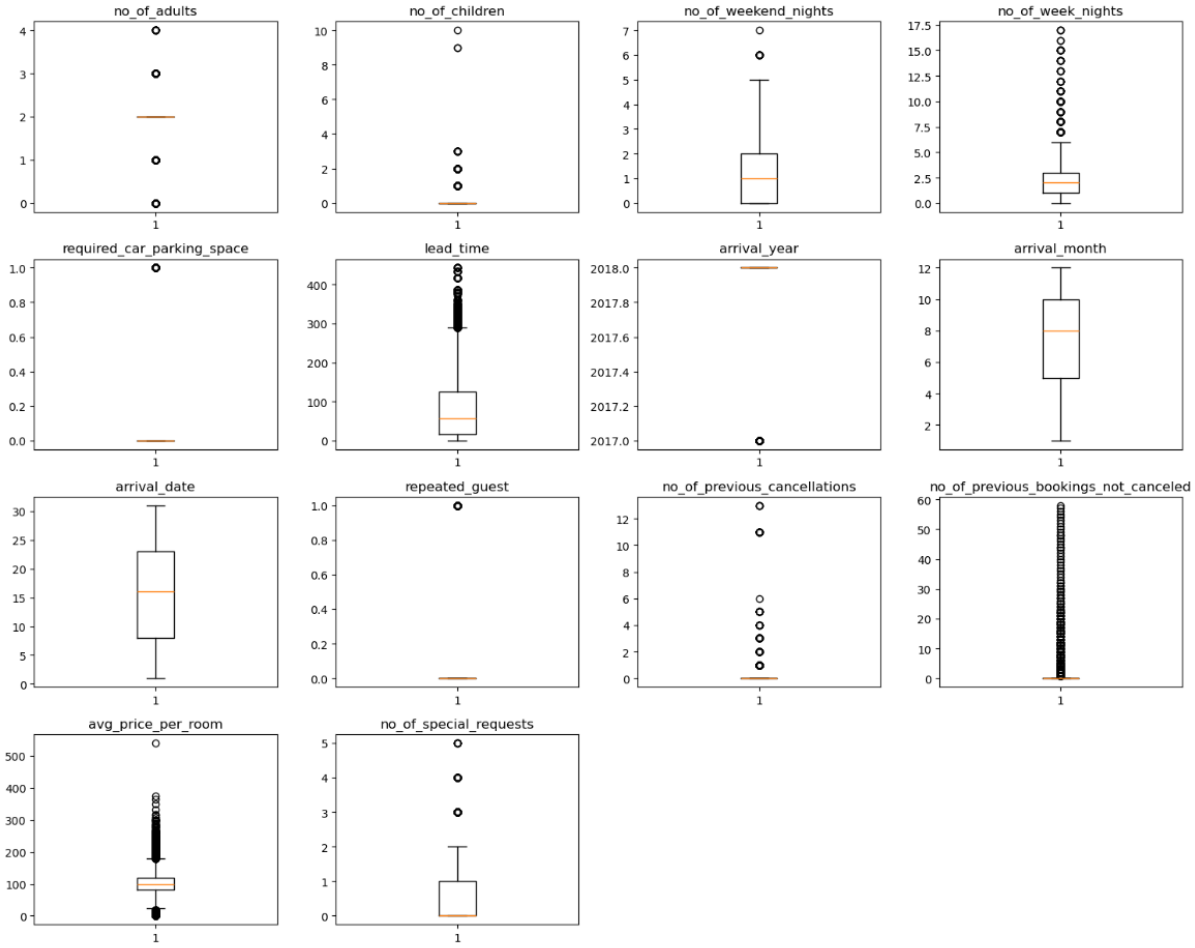


Figure 33: Boxplot of Outliers

Each of the numerical attributes contain outliers but these are real values which are true in nature in Realtime. So, treating them would be manipulating the data which in turn effects the model.

8.0 Data Preparation for modelling

```
Shape of Training set :  (25392, 27)
Shape of test set :  (10883, 27)
Shape of Training set :  (25392, 1)
Shape of test set :  (10883, 1)
Percentage of classes in training set:
Not_Canceled
1.00000        0.67399
0.00000        0.32601
Name: proportion, dtype: float64
Percentage of classes in test set:
Not_Canceled
1.00000        0.66857
0.00000        0.33143
Name: proportion, dtype: float64
```

8.1 Scaling the Data

	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	required_car_parking_space	lead_time	arrival_year	arrival_month	arrival_date	repeated_gu
0	0.29850	-0.26164	0.21880	0.57188	-0.17990	1.33640	0.46936	0.18828	1.53204	-0.16
1	0.29850	-0.26164	0.21880	0.57188	-0.17990	-0.07477	0.46936	-1.44595	0.95931	-0.16
2	0.29850	-0.26164	0.21880	1.28792	-0.17990	-0.08644	0.46936	-1.11911	-1.10252	-0.16
3	0.29850	-0.26164	1.37121	-1.57623	-0.17990	-0.28470	-2.13056	0.84197	-1.33161	-0.16
4	0.29850	2.22659	-0.93361	1.28792	-0.17990	1.34806	0.46936	1.16882	-1.67525	-0.16

Figure 34: Top 5 Rows of Scaled Train data

	no_of_adults	no_of_children	no_of_weekend_nights	no_of_week_nights	required_car_parking_space	lead_time	arrival_year	arrival_month	arrival_date	repeated_gu
0	0.29850	2.22659	0.21880	-1.57623	-0.17990	-0.96113	-2.13056	0.18828	0.84476	-0.16
1	0.29850	-0.26164	0.21880	-0.14416	-0.17990	-0.89116	0.46936	-1.77280	-0.41524	-0.16
2	0.29850	-0.26164	1.37121	-0.14416	-0.17990	-0.71622	0.46936	1.49566	1.07385	-0.16
3	0.29850	-0.26164	-0.93361	0.57188	-0.17990	-0.72788	0.46936	-0.46541	0.61567	-0.16
4	0.29850	2.22659	-0.93361	-0.14416	-0.17990	-0.89116	0.46936	0.51513	-0.87343	-0.16

Figure 35: Top 5 Rows of Scaled Test data

8.2 Logistic Regression (with Statsmodel)

```
                           Logit Regression Results
==============================================================================
Dep. Variable:           Not_Canceled   No. Observations:                25392
Model:                          Logit   Df Residuals:                    25364
Method:                           MLE   Df Model:                           27
Date:                Sat, 18 Jan 2025   Pseudo R-squ.:                  0.3273
Time:                        15:16:23   Log-Likelihood:                -10783.
converged:                       False   LL-Null:                       -16030.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==========================================================================================
                                          coef    std err          z      P>|z|      [0.025      0.975]
------------------------------------------------------------------------------------------
const                                    1.5417    229.175      0.007      0.995    -447.633     450.717
no_of_adults                            -0.0631      0.020     -3.231      0.001      -0.101      -0.025
no_of_children                          -0.0429      0.024     -1.799      0.072      -0.090       0.004
no_of_weekend_nights                    -0.1049      0.017     -6.095      0.000      -0.139      -0.071
no_of_week_nights                       -0.0447      0.017     -2.601      0.009      -0.078      -0.011
required_car_parking_space               0.2961      0.025     11.932      0.000       0.247       0.345
lead_time                               -1.3259      0.023    -58.373      0.000      -1.370      -1.281
arrival_year                            -0.1590      0.023     -6.945      0.000      -0.204      -0.114
arrival_month                            0.1310      0.020      6.592      0.000       0.092       0.170
arrival_date                            -0.0224      0.017     -1.326      0.185      -0.056       0.011
repeated_guest                           0.4028      0.103      3.908      0.000       0.201       0.605
no_of_previous_cancellations            -0.1008      0.029     -3.431      0.001      -0.158      -0.043
no_of_previous_bookings_not_canceled     0.1294      0.161      0.803      0.422      -0.186       0.445
avg_price_per_room                      -0.6538      0.026    -25.313      0.000      -0.704      -0.603
no_of_special_requests                   1.1687      0.024     49.096      0.000       1.122       1.215
type_of_meal_plan_Meal Plan 2           -0.0514      0.019     -2.687      0.007      -0.089      -0.014
type_of_meal_plan_Meal Plan 3           -0.1772    134.628     -0.001      0.999    -264.044     263.690
type_of_meal_plan_Not Selected          -0.0667      0.018     -3.615      0.000      -0.103      -0.031
room_type_reserved_Room_Type 2           0.0602      0.018      3.325      0.001       0.025       0.096
room_type_reserved_Room_Type 3          -0.0138      0.025     -0.555      0.579      -0.063       0.035
room_type_reserved_Room_Type 4           0.1065      0.020      5.370      0.000       0.068       0.145
room_type_reserved_Room_Type 5           0.0585      0.018      3.309      0.001       0.024       0.093
room_type_reserved_Room_Type 6           0.1492      0.024      6.205      0.000       0.102       0.196
room_type_reserved_Room_Type 7           0.0816      0.020      4.069      0.000       0.042       0.121
market_segment_type_Complementary        3.4873   2210.479      0.002      0.999   -4328.973    4335.947
market_segment_type_Corporate            0.1971      0.063      3.125      0.002       0.073       0.321
market_segment_type_Offline              0.8349      0.119      6.992      0.000       0.601       1.069
market_segment_type_Online               0.0219      0.125      0.175      0.861      -0.223       0.266
==========================================================================================
```

Figure 36: Logistic Regression (with Statsmodel)

8.2.1 Checking Logistic Regression model performance on training set

	Accuracy	Recall	Precision	F1
0	0.80502	0.89079	0.83183	0.86030

Figure 37: Performance of model on Training set

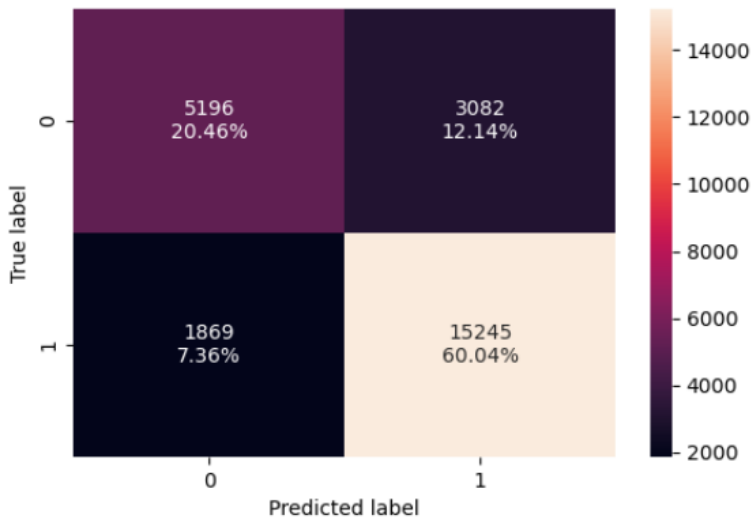


Figure 38: Confusion Matrix

8.2.1 Checking Logistic Regression model performance on test set

	Accuracy	Recall	Precision	F1
0	0.80722	0.89788	0.82822	0.86165

Figure 39: Performance of model on test set

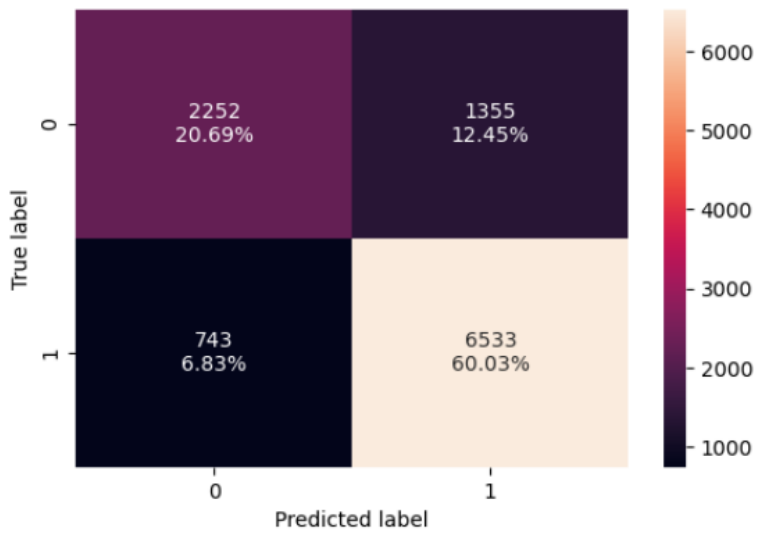


Figure 40: Confusion matrix

8.3 Naive - Bayes Classifier

8.3.1 Checking Naive - Bayes Classifier performance on training set

	Accuracy	Recall	Precision	F1
0	0.40580	0.13404	0.89539	0.23318

Figure 41: Performance of model on train data

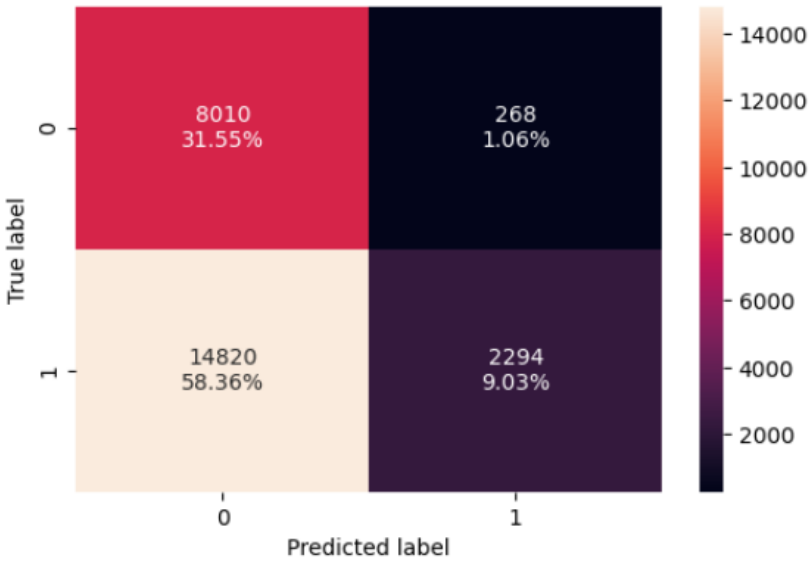


Figure 42: Confusion matrix

8.3.2 Checking Naive - Bayes Classifier performance on test set

	Accuracy	Recall	Precision	F1
0	0.41055	0.13386	0.89604	0.23293

Figure 43: Performance of model on test set

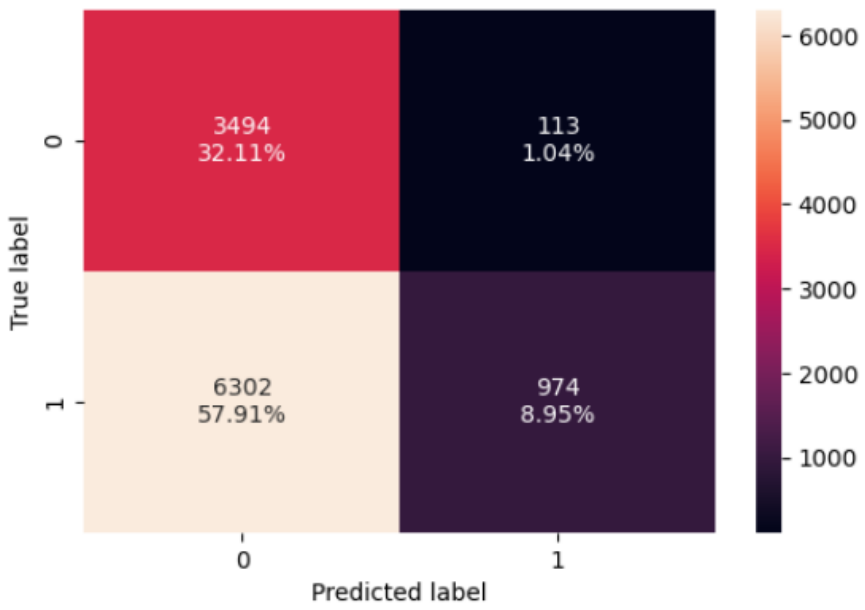


Figure 44: Confusion matrix

8.4 KNN Classifier (K = 3)

8.4.1 Checking KNN Classifier performance on training set

	Accuracy	Recall	Precision	F1
0	0.91332	0.94589	0.92699	0.93634

Figure 45: Performance of model on training set

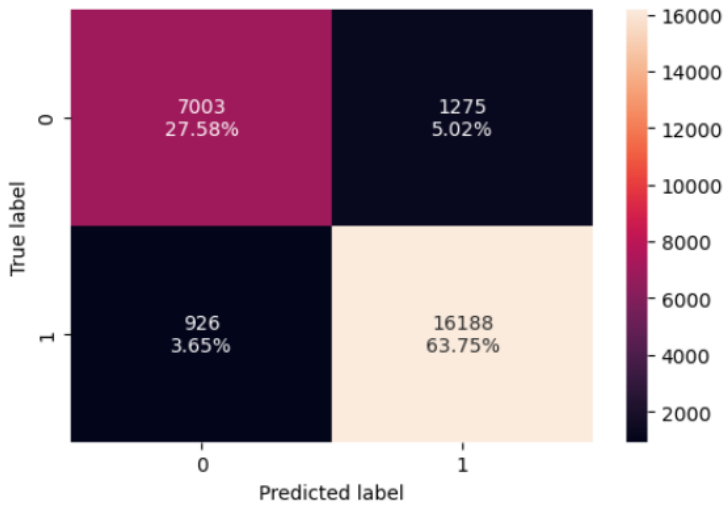


Figure 46: Confusion Matrix

8.4.2 Checking KNN Classifier performance on test set

	Accuracy	Recall	Precision	F1
0	0.84793	0.89843	0.87710	0.88764

Figure 47: Performance on test set

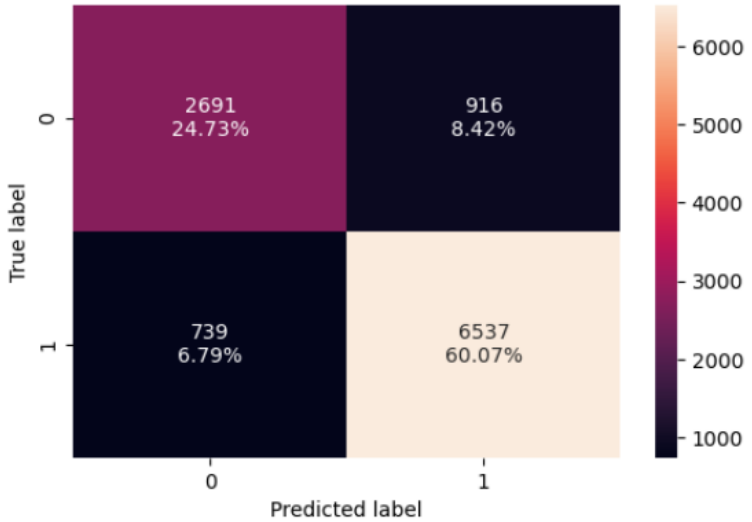


Figure 48: Confusion Matrix

8.5 Decision Tree Classifier

8.5.1 Checking Decision Tree Classifier performance on training set

	Accuracy	Recall	Precision	F1
0	0.65060	0.95659	0.66820	0.78680

Figure 49: Performance of model on training set

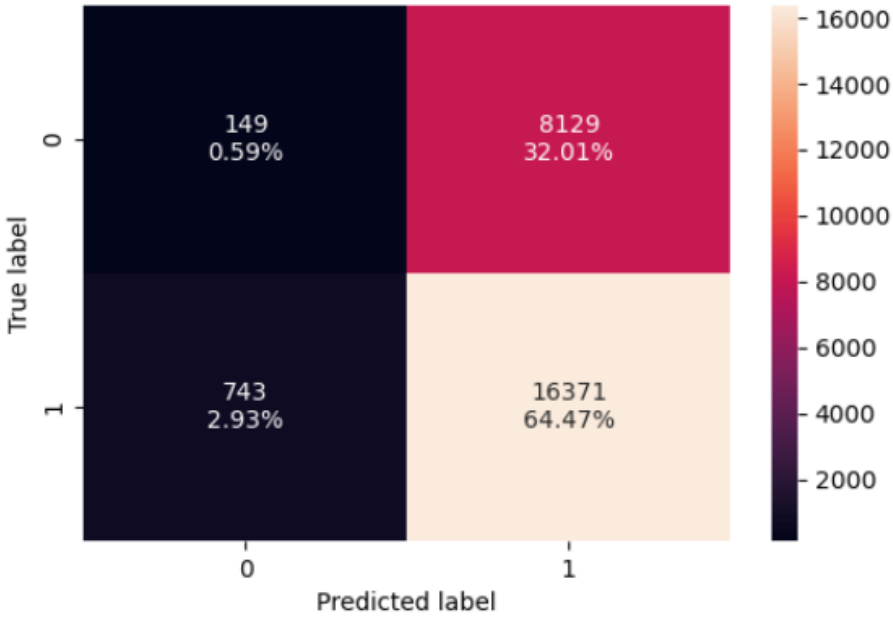


Figure 50: Confusion matrix

8.5.2 Checking Decision Tree Classifier performance on test set

	Accuracy	Recall	Precision	F1
0	0.64440	0.95602	0.66210	0.78236

Figure 51: Performance of model on test set

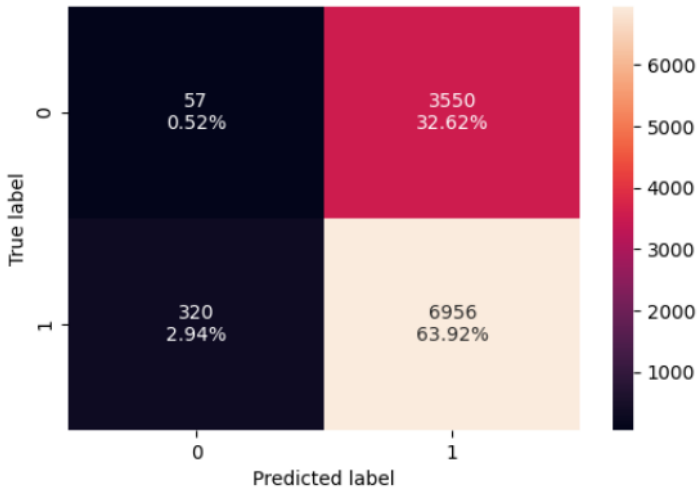


Figure 52: Confusion matrix

9.0 Model Performance Improvement

Logistic Regression (deal with multicollinearity, remove high p-value variables, determine optimal threshold using ROC curve)

```
Variance Inflation Factors:
                                Variable      VIF
0                                  const  1.00000
1                           no_of_adults  1.34299
2                          no_of_children  2.00637
3                    no_of_weekend_nights  1.06511
4                       no_of_week_nights  1.09149
5              required_car_parking_space  1.03707
6                              lead_time  1.39281
7                           arrival_year  1.43007
8                          arrival_month  1.27171
9                           arrival_date  1.00666
10                         repeated_guest  1.76765
11           no_of_previous_cancellations  1.37023
12  no_of_previous_bookings_not_canceled  1.61806
13                     avg_price_per_room  2.04856
14                 no_of_special_requests  1.24585
15          type_of_meal_plan_Meal Plan 2  1.26460
16          type_of_meal_plan_Meal Plan 3  1.00610
17         type_of_meal_plan_Not Selected  1.27551
18         room_type_reserved_Room_Type 2  1.09146
19         room_type_reserved_Room_Type 3  1.00490
20         room_type_reserved_Room_Type 4  1.35628
21         room_type_reserved_Room_Type 5  1.03320
22         room_type_reserved_Room_Type 6  1.98810
23         room_type_reserved_Room_Type 7  1.10446
24      market_segment_type_Complementary  4.49253
25          market_segment_type_Corporate 17.21348
26            market_segment_type_Offline 64.41691
27             market_segment_type_Online 71.43903
```

Figure 53: Attributes

Dropping columns with VIF > 5 iteratively

```
Dropping market_segment_type_Online due to high VIF
```

Figure 54: Dropped Attribute

	Variable	VIF
0	no_of_adults	1.32591
1	no_of_children	2.00540
2	no_of_weekend_nights	1.06439
3	no_of_week_nights	1.09028
4	required_car_parking_space	1.03704
5	lead_time	1.38817
6	arrival_year	1.42771
7	arrival_month	1.27096
8	arrival_date	1.00664
9	repeated_guest	1.76569
10	no_of_previous_cancellations	1.37006
11	no_of_previous_bookings_not_canceled	1.61793
12	avg_price_per_room	2.04799
13	no_of_special_requests	1.24130
14	type_of_meal_plan_Meal Plan 2	1.26427
15	type_of_meal_plan_Meal Plan 3	1.00610
16	type_of_meal_plan_Not Selected	1.27359
17	room_type_reserved_Room_Type 2	1.09127
18	room_type_reserved_Room_Type 3	1.00490
19	room_type_reserved_Room_Type 4	1.35202
20	room_type_reserved_Room_Type 5	1.03320
21	room_type_reserved_Room_Type 6	1.98782
22	room_type_reserved_Room_Type 7	1.10434
23	market_segment_type_Complementary	1.33887
24	market_segment_type_Corporate	1.53973
25	market_segment_type_Offline	1.60447

Figure 55: Attributes and their VIFs

9.1 Dealing with high p-value variables

```
Warning: Maximum number of iterations has been exceeded.
         Current function value: 0.424659
         Iterations: 35
Dropping column type_of_meal_plan_Meal Plan 3 with p-value: 0.9989496591537229
Warning: Maximum number of iterations has been exceeded.
         Current function value: 0.424703
         Iterations: 35
Dropping column market_segment_type_Complementary with p-value: 0.9998497550672123
Optimization terminated successfully.
         Current function value: 0.425167
         Iterations 10
Dropping column room_type_reserved_Room_Type 3 with p-value: 0.6591034670615359
Optimization terminated successfully.
         Current function value: 0.425171
         Iterations 10
Dropping column no_of_previous_bookings_not_canceled with p-value: 0.419072300250489
Optimization terminated successfully.
         Current function value: 0.425190
         Iterations 9
Dropping column arrival_date with p-value: 0.19830155371616764
Optimization terminated successfully.
         Current function value: 0.425223
         Iterations 9
Dropping column market_segment_type_Corporate with p-value: 0.16086258278487142
Optimization terminated successfully.
         Current function value: 0.425260
         Iterations 9
Dropping column no_of_children with p-value: 0.07881927209324384
Optimization terminated successfully.
         Current function value: 0.425319
         Iterations 9
Dropping column type_of_meal_plan_Meal Plan 2 with p-value: 0.009443733608939412
['const', 'no_of_adults', 'no_of_weekend_nights', 'no_of_week_nights', 'required_car_parking_space', 'lead_time', 'arrival_year', 'arrival_month', 're
peated_guest', 'no_of_previous_cancellations', 'avg_price_per_room', 'no_of_special_requests', 'type_of_meal_plan_Meal Plan 2', 'type_of_meal_plan_Not
Selected', 'room_type_reserved_Room_Type 2', 'room_type_reserved_Room_Type 4', 'room_type_reserved_Room_Type 5', 'room_type_reserved_Room_Type 6', 'ro
om_type_reserved_Room_Type 7', 'market_segment_type_Offline', 'market_segment_type_Online']
```

Figure 56: P-value of the variables

After dropping variables with high p-value variables and here are the significant variables.

	const	no_of_adults	no_of_weekend_nights	no_of_week_nights	required_car_parking_space	lead_time	arrival_year	arrival_month	repeated_guest	no_of_previous_
0	1.00000	0.29850	0.21880	0.57188	-0.17990	1.33640	0.46936	0.18828	-0.16067	
1	1.00000	0.29850	0.21880	0.57188	-0.17990	-0.07477	0.46936	-1.44595	-0.16067	
2	1.00000	0.29850	0.21880	1.28792	-0.17990	-0.08644	0.46936	-1.11911	-0.16067	
3	1.00000	0.29850	1.37121	-1.57623	-0.17990	-0.28470	-2.13056	0.84197	-0.16067	
4	1.00000	0.29850	-0.93361	1.28792	-0.17990	1.34806	0.46936	1.16882	-0.16067	
5	1.00000	0.29850	-0.93361	-0.14416	-0.17990	-0.85617	0.46936	-1.77280	-0.16067	
6	1.00000	0.29850	-0.93361	-0.86019	-0.17990	-0.90282	0.46936	-1.77280	-0.16067	
7	1.00000	-1.62816	0.21880	-0.86019	-0.17990	2.51432	0.46936	-0.13857	-0.16067	
8	1.00000	-1.62816	0.21880	-0.14416	-0.17990	-0.26138	0.46936	-0.79226	-0.16067	
9	1.00000	0.29850	0.21880	-0.14416	-0.17990	-0.33135	0.46936	-0.46541	-0.16067	

Figure 57: Significant variables

9.2 Training the Logistic Regression model again with only the significant features

```
Optimization terminated successfully.
         Current function value: 0.425319
         Iterations 9
                           Logit Regression Results
==============================================================================
Dep. Variable:           Not_Canceled   No. Observations:                25392
Model:                          Logit   Df Residuals:                    25371
Method:                           MLE   Df Model:                           20
Date:                Sat, 18 Jan 2025   Pseudo R-squ.:                  0.3263
Time:                        15:16:36   Log-Likelihood:                -10800.
converged:                       True   LL-Null:                       -16030.
Covariance Type:            nonrobust   LLR p-value:                     0.000
==================================================================================================
                                     coef    std err          z      P>|z|      [0.025      0.975]
--------------------------------------------------------------------------------------------------
const                              1.1866      0.023     50.623      0.000       1.141       1.233
no_of_adults                      -0.0557      0.019     -2.881      0.004      -0.094      -0.018
no_of_weekend_nights              -0.1087      0.017     -6.330      0.000      -0.142      -0.075
no_of_week_nights                 -0.0492      0.017     -2.863      0.004      -0.083      -0.016
required_car_parking_space         0.2957      0.025     11.926      0.000       0.247       0.344
lead_time                         -1.3237      0.023    -58.439      0.000      -1.368      -1.279
arrival_year                      -0.1599      0.023     -7.008      0.000      -0.205      -0.115
arrival_month                      0.1338      0.020      6.748      0.000       0.095       0.173
repeated_guest                     0.4443      0.096      4.623      0.000       0.256       0.633
no_of_previous_cancellations      -0.0971      0.028     -3.411      0.001      -0.153      -0.041
avg_price_per_room                -0.6681      0.025    -26.272      0.000      -0.718      -0.618
no_of_special_requests             1.1664      0.024     49.137      0.000       1.120       1.213
type_of_meal_plan_Meal Plan 2     -0.0496      0.019     -2.596      0.009      -0.087      -0.012
type_of_meal_plan_Not Selected    -0.0663      0.018     -3.605      0.000      -0.102      -0.030
room_type_reserved_Room_Type 2     0.0521      0.018      2.958      0.003       0.018       0.087
room_type_reserved_Room_Type 4     0.1075      0.020      5.450      0.000       0.069       0.146
room_type_reserved_Room_Type 5     0.0605      0.018      3.431      0.001       0.026       0.095
room_type_reserved_Room_Type 6     0.1251      0.019      6.594      0.000       0.088       0.162
room_type_reserved_Room_Type 7     0.0776      0.020      3.937      0.000       0.039       0.116
market_segment_type_Offline        0.4634      0.046     10.139      0.000       0.374       0.553
market_segment_type_Online        -0.3699      0.046     -7.972      0.000      -0.461      -0.279
==================================================================================================
```

Figure 58: Logistic model summary after tuning

9.3 Determining optimal threshold using ROC Curve

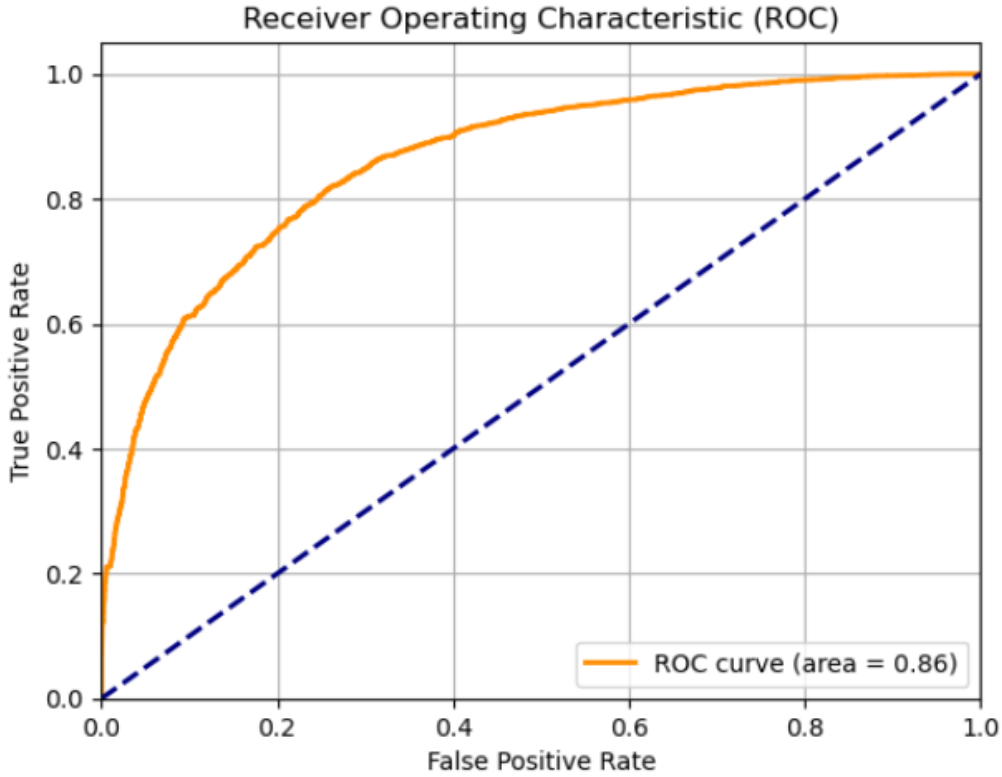


Figure 59: ROC Curve

the ROC curve analysis indicates a well-performing model, and the AUC of 0.86 reflects its reliability in classification tasks.

```
Optimal Threshold:  0.637
```

Figure 60: Optimal Threshold

9.4 Checking tuned Logistic Regression model performance on training set.

	Accuracy	Recall	Precision	F1
0	0.79214	0.81647	0.86735	0.84114

Figure 61: Performance of Logistic Regression model on Training set

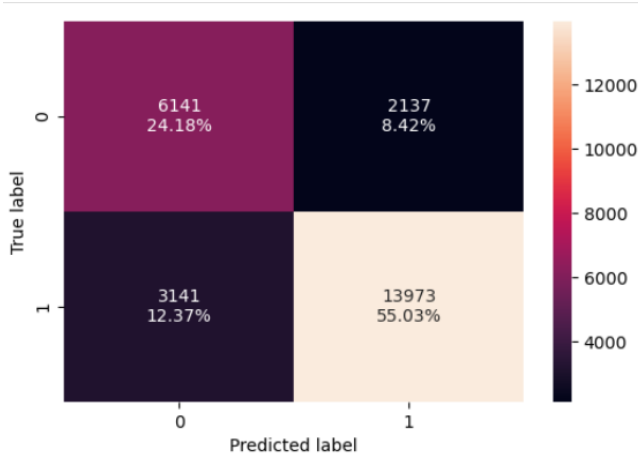


Figure 62:Confusion Matrix

9.5 Checking tuned Logistic Regression model performance on test set.

	Accuracy	Recall	Precision	F1
0	0.79261	0.82614	0.85835	0.84194

Figure 63: Performance of tuned model on test set

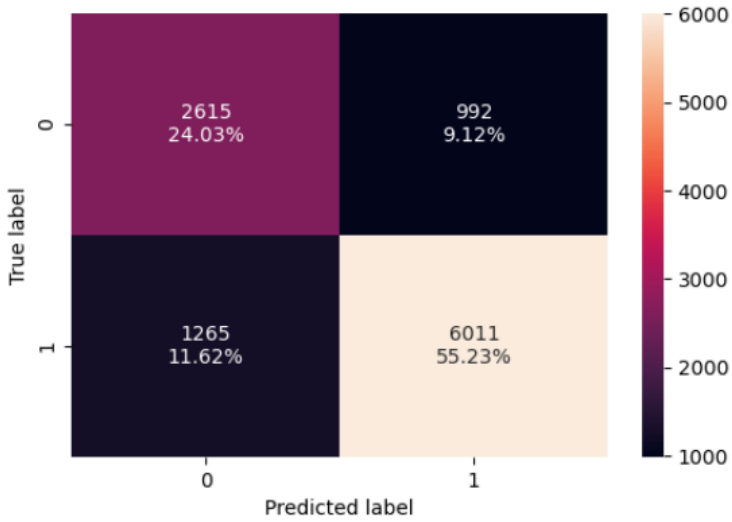


Figure 64: Confusion matrix

9.6 KNN Classifier (different values of K)

```
Recall for k=2: 0.7978284771852666
Recall for k=3: 0.8982957669048928
Recall for k=4: 0.8519791094007697
Recall for k=5: 0.9043430456294668
Recall for k=6: 0.8714953271028038
Recall for k=7: 0.9029686641011545
Recall for k=8: 0.8815283122594832
Recall for k=9: 0.9061297416162727
Recall for k=10: 0.8862012094557449
Recall for k=11: 0.9059923034634415
Recall for k=12: 0.8867509620670698
Recall for k=13: 0.9050302363936229
Recall for k=14: 0.8892248488180319
Recall for k=15: 0.90695437053326
Recall for k=16: 0.8925233644859814
Recall for k=17: 0.9087410665200659
Recall for k=18: 0.8962341946124244
Recall for k=19: 0.9113523914238593
Recall for k=20: 0.8989829576690489

The best value of k is: 19 with a recall of: 0.9113523914238593
```

Figure 65: KNN Classifier with different K values

9.7 Checking tuned KNN model performance on training set.

	Accuracy	Recall	Precision	F1
0	0.85425	0.91469	0.87477	0.89428

Figure 66: Model performance on train set

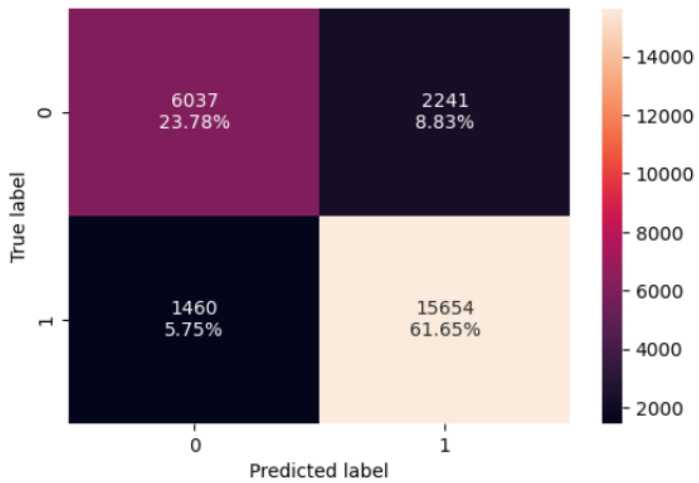


Figure 67: Confusion matrix

9.8 Checking tuned KNN model performance on test set.

	Accuracy	Recall	Precision	F1
0	0.84333	0.91135	0.86218	0.88608

Figure 68: Model performance on test set

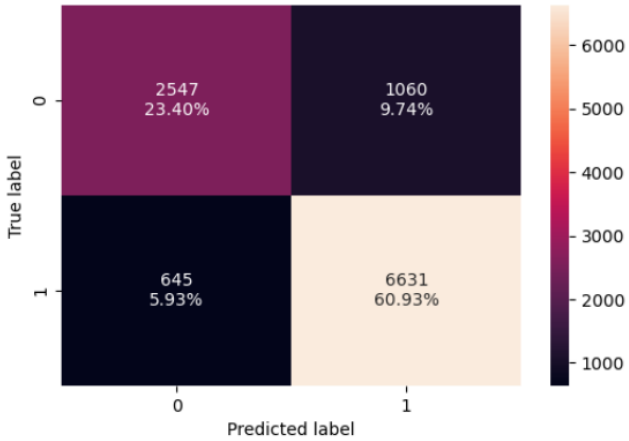


Figure 69: Confusion Matrix

9.9 Decision Tree Classifier (pre-pruning)

Max_depth = 11

Max_leaf_nodes =100

Min_samples_split = 30

Random_state = 42

9.10 Checking tuned Decision Tree Classifier performance on training set

	Accuracy	Recall	Precision	F1
0	0.87571	0.92906	0.89116	0.90972

Figure 70: Model performance on train set

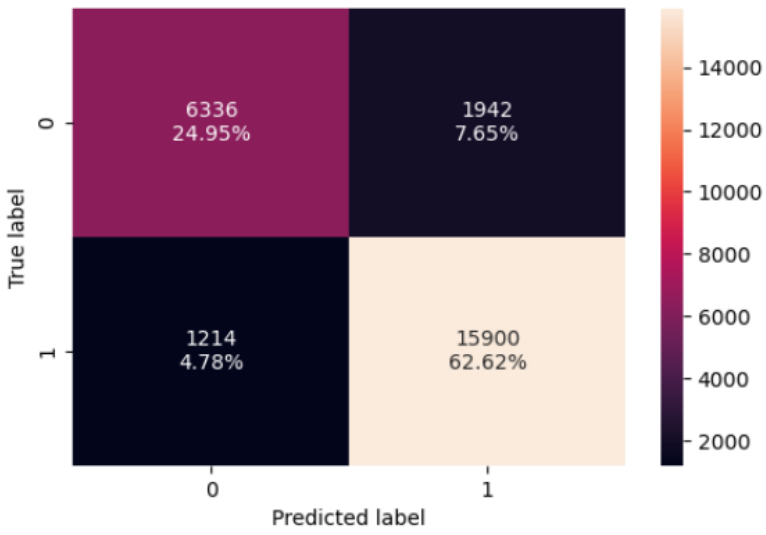


Figure 71: Confusion Matrix

9.11 Checking tuned Decision Tree Classifier performance on test set

	Accuracy	Recall	Precision	F1
0	0.87384	0.93073	0.88627	0.90796

Figure 72: Model Performance on test set

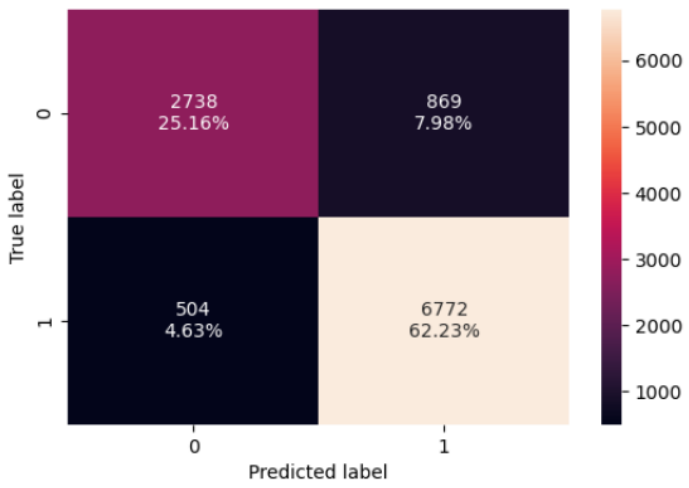


Figure 73: Confusion Matrix

9.12 Visualizing the Decision Tree

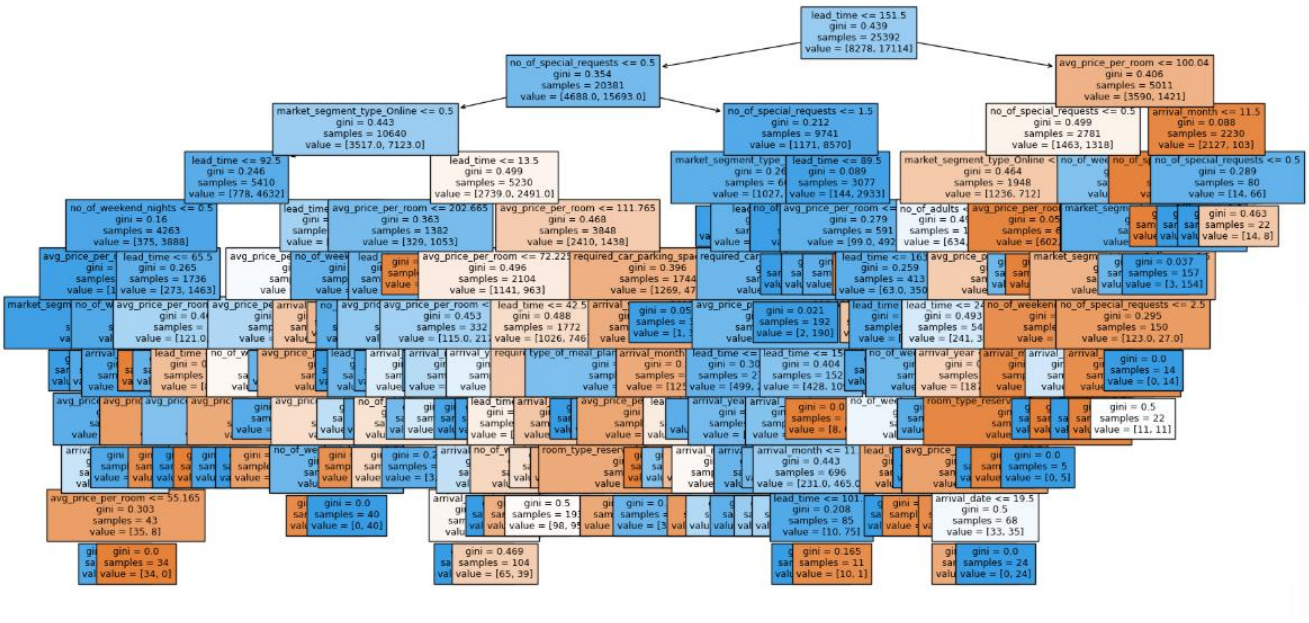


Figure 74: Visualizing the Decision Tree

9.12 Analysing Feature Importance for tuned Decision Tree Classifier

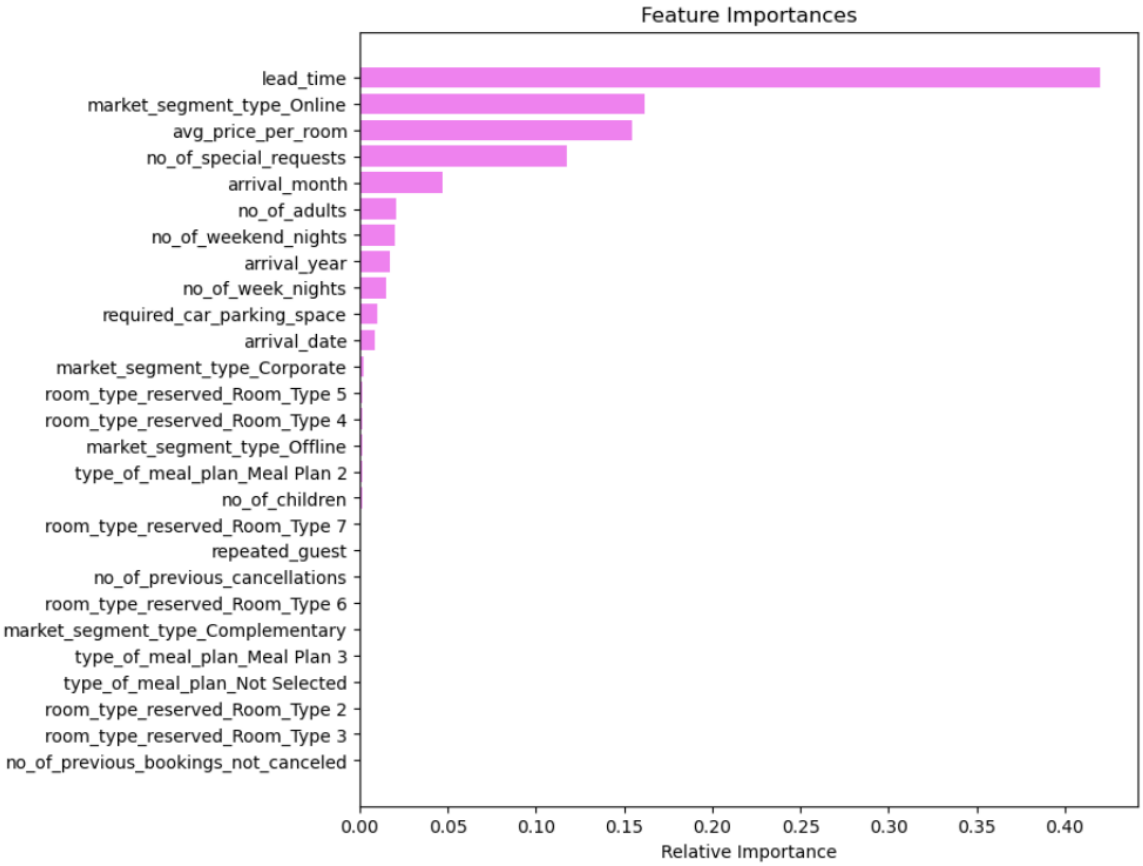


Figure 75: Feature Importance for tuned Decision Tree Classifier

10.0 Model Performance Comparison and Final Model Selection

Training performance comparison:

	Logistic Regression Base	Logistic Regression Tuned	Naive Bayes Base	KNN Base	KNN Tuned	Decision Tree Base	Decision Tree Tuned
Accuracy	0.80502	0.79214	0.40580	0.91332	0.85425	0.65060	0.87571
Recall	0.89079	0.81647	0.13404	0.94589	0.91469	0.95659	0.92906
Precision	0.83183	0.86735	0.89539	0.92699	0.87477	0.66820	0.89116
F1	0.86030	0.84114	0.23318	0.93634	0.89428	0.78680	0.90972

Figure 76: Model Performance Comparison and Final Model Selection on train set

Test set performance comparison:

	Logistic Regression Base	Logistic Regression Tuned	Naive Bayes Base	KNN Base	KNN Tuned	Decision Tree Base	Decision Tree Tuned
Accuracy	0.80502	0.79214	0.41055	0.84793	0.84333	0.64440	0.87384
Recall	0.89079	0.81647	0.13386	0.89843	0.91135	0.95602	0.93073
Precision	0.83183	0.86735	0.89604	0.87710	0.86218	0.66210	0.88627
F1	0.86030	0.84114	0.23293	0.88764	0.88608	0.78236	0.90796

Figure 77: Model Performance Comparison and Final Model Selection on test set

Decision Tree Tuned is the best model. As comparing with other models its performing well on this model.

11.0 Actionable Insights and Recommendations

Actionable Insights:

Lead Time:

Insight: Longer lead times significantly increase the likelihood of cancellations.

Recommendation: Consider implementing stricter cancellation policies for bookings made well in advance. For example, require a non-refundable deposit for long lead-time bookings.

Market Segment Type (Online):

Insight: Bookings made through online channels have a higher likelihood of cancellation.

Recommendation: Offer incentives such as discounts or loyalty points for customers who book directly through the hotel's website or app. This can help reduce dependency on third-party platforms and potentially lower cancellation rates.

Average Price per Room:

Insight: Higher-priced rooms are more likely to be cancelled.

Recommendation: Implement flexible pricing strategies to encourage customers to commit to their reservations. For example, offer special non-refundable rates with discounts for higher-priced rooms.

Number of Special Requests:

Insight: Guests who make multiple special requests are more likely to cancel.

Recommendation: Track and review special requests to identify patterns and trends. Consider reaching out to these guests with personalized communication to confirm their reservations and address their needs.

Recommendations:

Implement Tiered Cancellation Policies:

Develop tiered cancellation policies based on booking characteristics such as lead time, room type, and booking channel. For example, offer flexible cancellation options for last-minute bookings but enforce stricter policies for long lead-time bookings.

Optimize Distribution Channels:

Analyse the performance of various distribution channels and optimize the mix to reduce cancellations. Prioritize channels that have lower cancellation rates and offer incentives for direct bookings.

Enhance Customer Communication:

Improve pre-stay communication with guests to reduce cancellations. Send reminders, special offers, and personalized messages to keep guests engaged and committed to their reservations.

Offer Incentives for Non-Cancellation:

Introduce loyalty programs or incentives for guests who honour their bookings. Offer discounts, loyalty points, or other rewards for guests who consistently avoid cancellations.

Monitor and Adjust Pricing Strategies:

Use dynamic pricing strategies to adjust room rates based on booking patterns and cancellation likelihood. Offer special non-refundable rates or discounts for guests who are less likely to cancel.

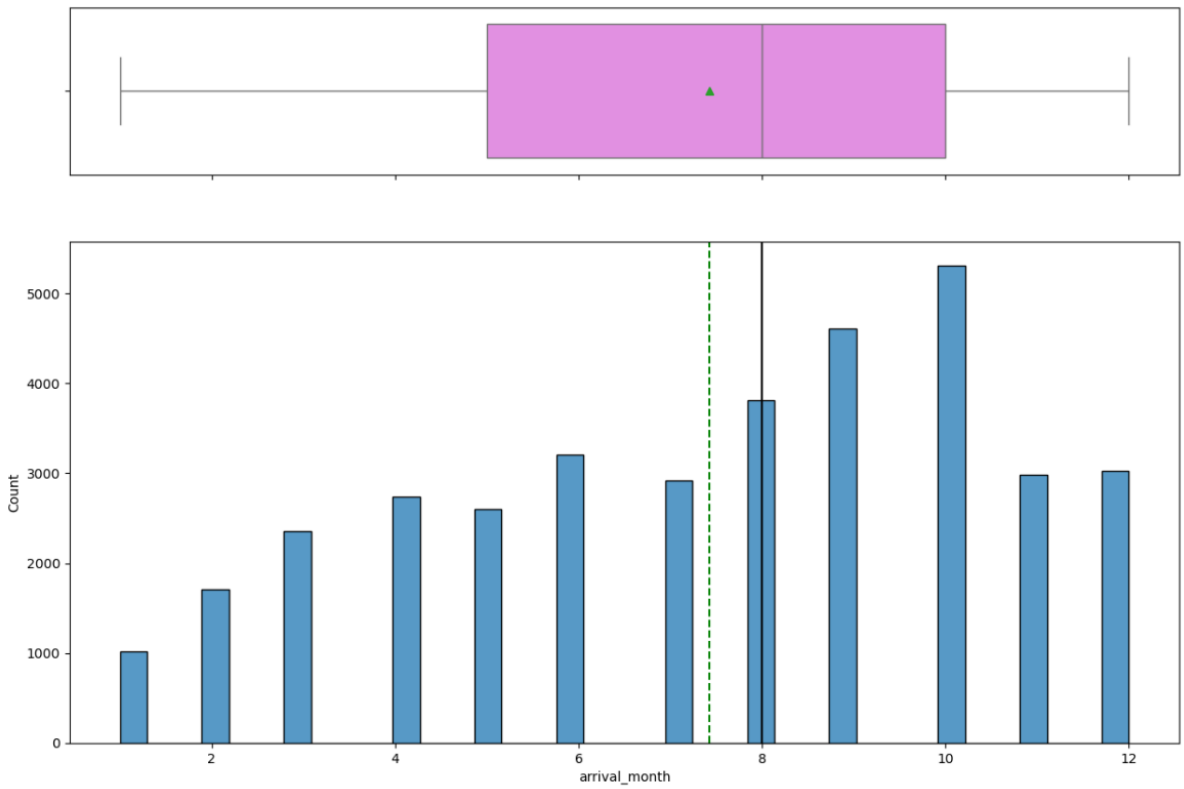
Review and Refine Special Requests Management:

Monitor special requests closely and identify guests with high cancellation risk. Offer personalized services and communication to address their needs and reduce the likelihood of cancellations.

By implementing these insights and recommendations, INN Hotels Group can proactively manage booking cancellations, optimize their revenue, and improve guest satisfaction.

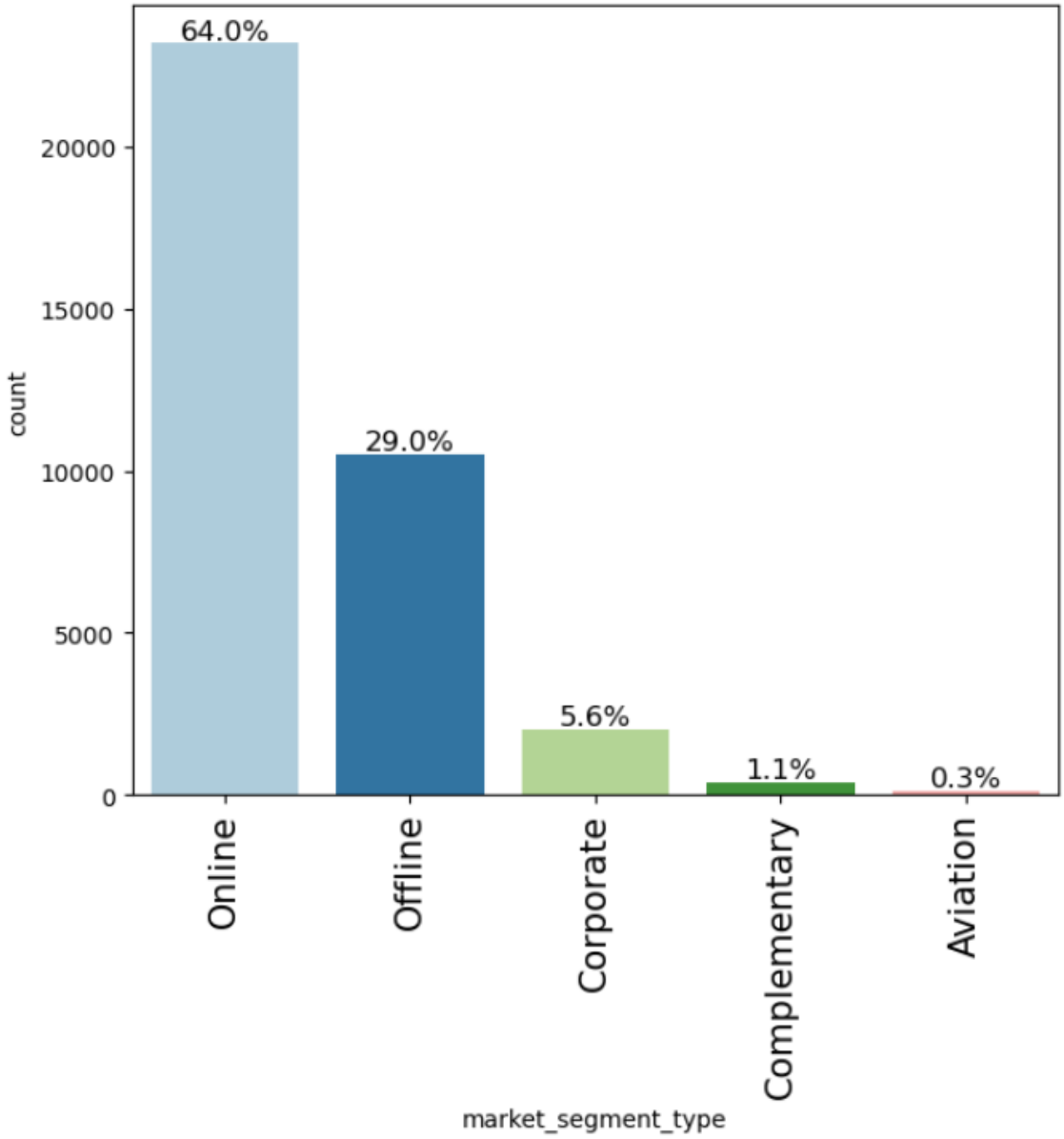
12.0 Answers to EDA questions

1. What are the busiest months in the hotel?



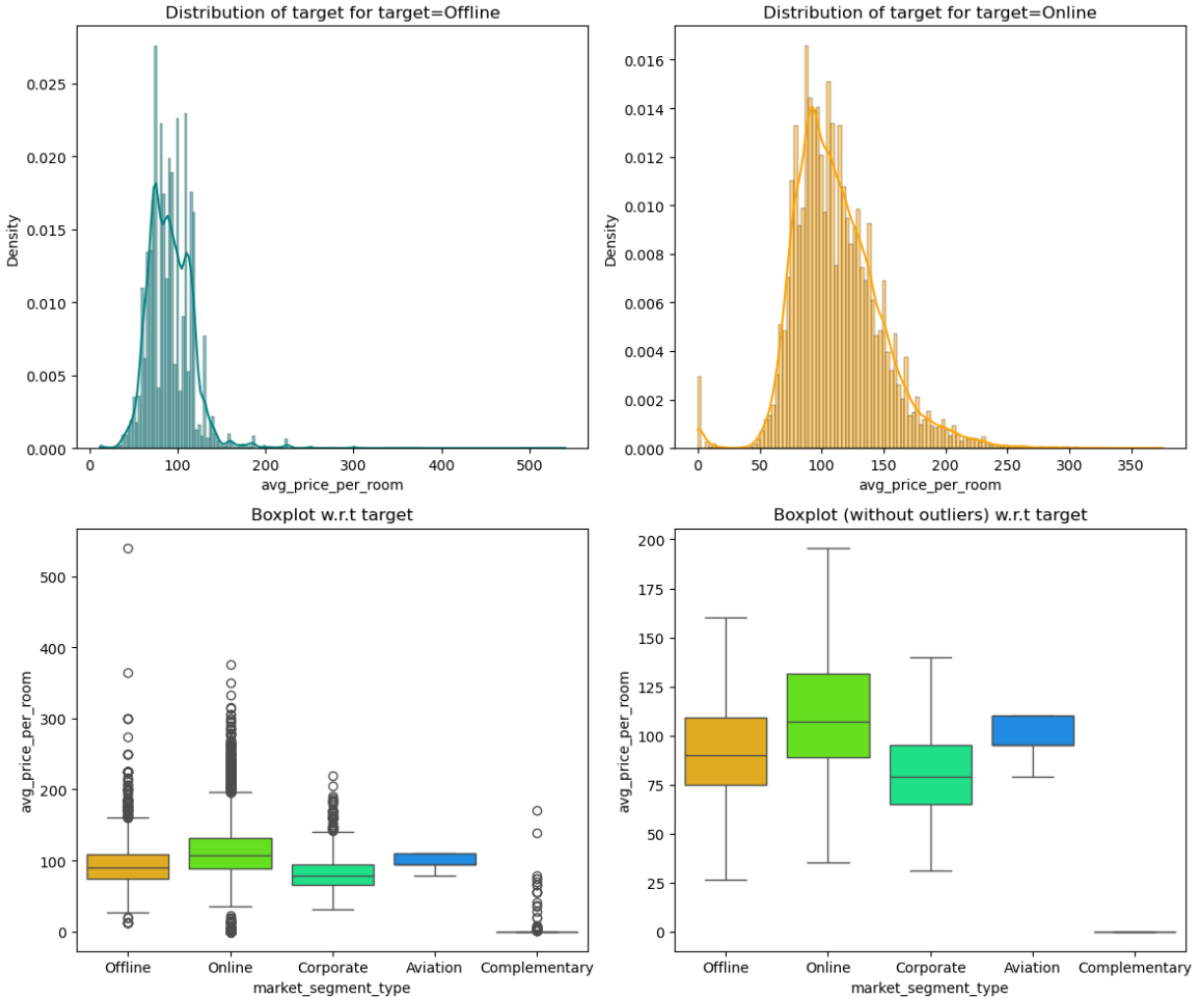
- Most of the bookings are in Month of October, September, and August. One can say these are the busiest months in the hotel.

2. Which market segment do most of the guests come from?



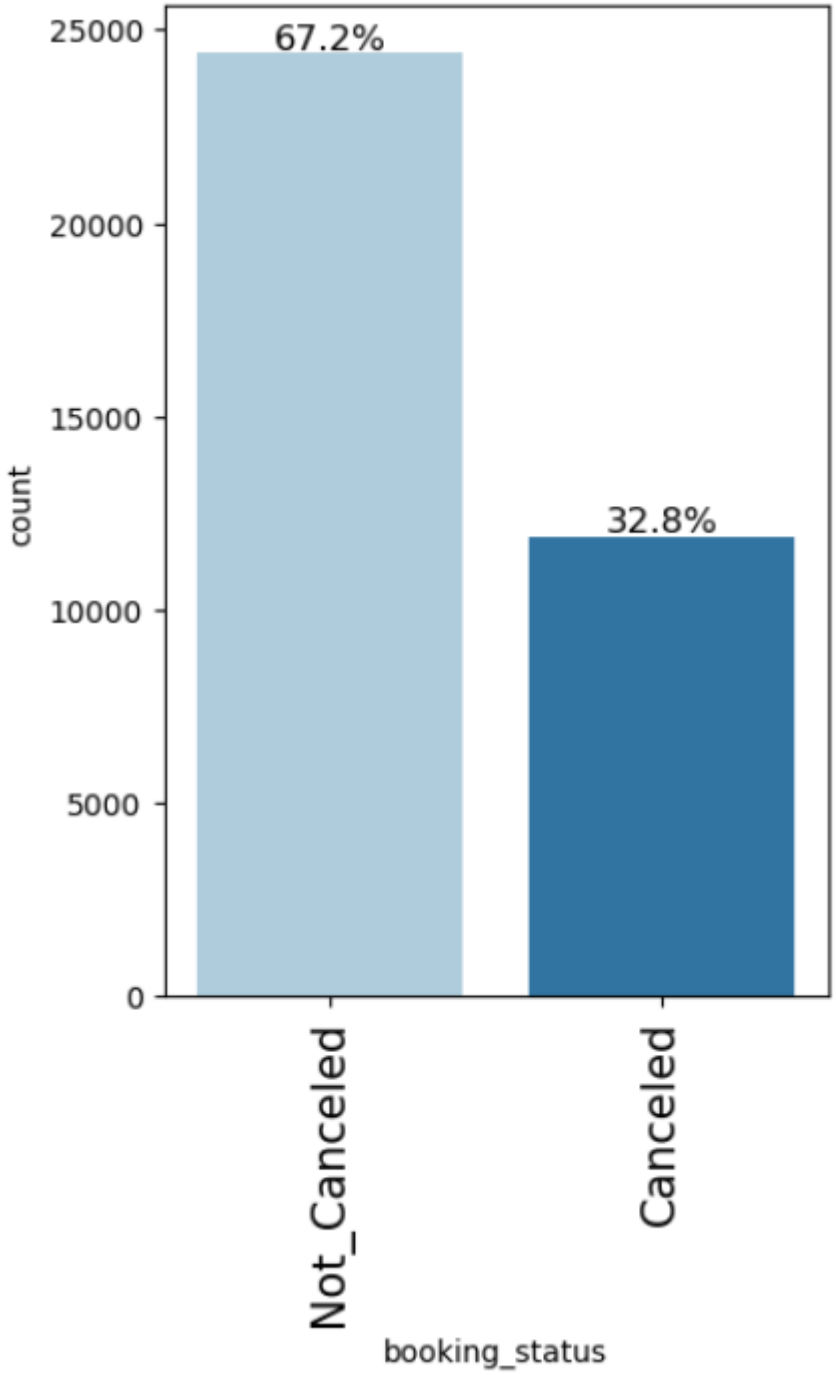
- Most of the bookings are of Online, then Offline.

3. Hotel rates are dynamic and change according to demand and customer demographics. What are the differences in room prices in different market segments?



- Avg. Price of Online Market Segment is high compared to other segments.

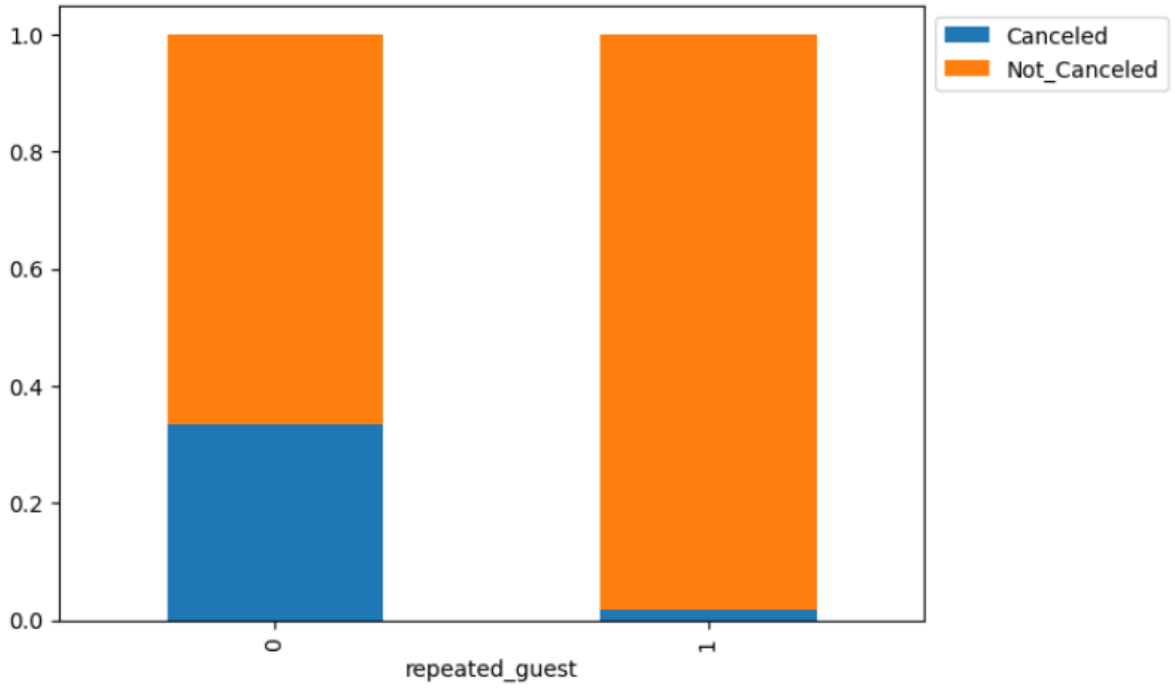
4. What percentage of bookings are cancelled?



- 32.8% of bookings are cancelled.

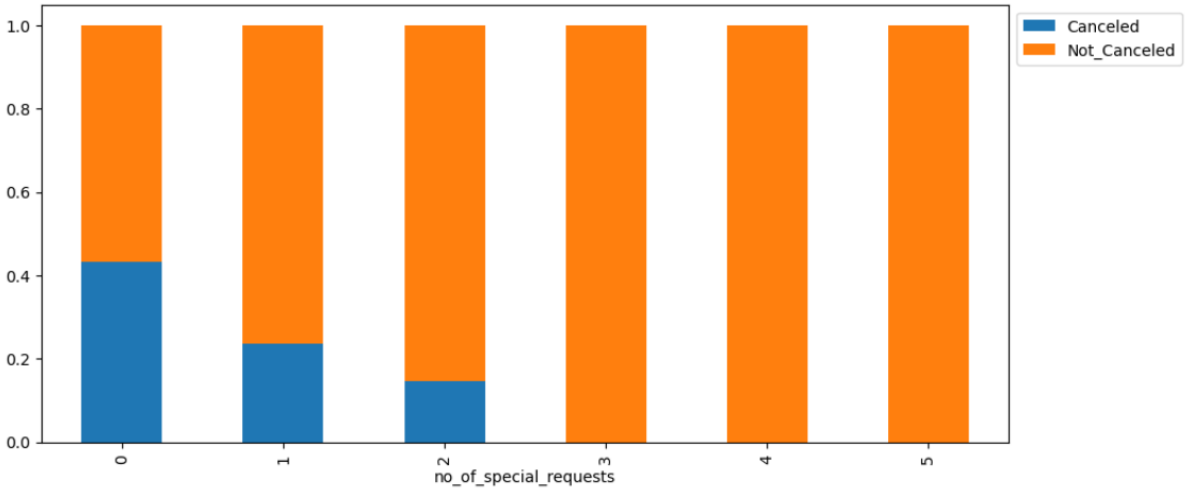
5. Repeating guests are the guests who stay in the hotel often and are important to brand equity. What percentage of repeating guests cancel?

booking_status	Canceled	Not_Canceled	All
repeated_guest			
All	11885	24390	36275
0	11869	23476	35345
1	16	914	930



6.0 Many guests have special requirements when booking a hotel room. Do these requirements affect booking cancellation?

booking_status	Canceled	Not_Canceled	All
no_of_special_requests			
All	11885	24390	36275
0	8545	11232	19777
1	2703	8670	11373
2	637	3727	4364
3	0	675	675
4	0	78	78
5	0	8	8



Yes. Requirements affect booking cancellation.